Null-Directional Kinematics and the Statistical Structure of Spacetime

Clock Functionals, Transport Geometry, and Directional Support

Eric Cole Thompson
Independent Researcher
Fulshear, TX, USA
ethompson06@gmail.com

Abstract

A minimal kinematic framework is investigated in which physical structure is organized by a normalized ensemble of invariant-speed null directions. Assuming null propagation as primitive, spatial direction is encoded by a probability distribution on the celestial sphere. Within this setting it is shown that, in one spatial dimension, the proper-time rate of an idealized local clock is fixed by adopting the symmetric overlap of opposing directional weights, yielding the Lorentz factor without presupposing spacetime geometry.

The first moment of the null-directional distribution defines a transport field whose stationary, irrotational limit reproduces the Newtonian potential structure. At the quantum level, the Fourier duality between spatial localization and directional (momentum) resolution constrains the admissible state space, reproducing the standard Heisenberg uncertainty bound within this directional representation. Promoting deviations from isotropy to a covariant scalar strain functional leads to a compatibility closure in which null focusing and metric curvature align with the Einstein equations under standard locality and covariance assumptions.

Lifting directional amplitudes to their spinor representation introduces an intrinsic internal frame redundancy, from which gauge connections are recovered as the minimal structures required for local comparison. Finally, a Planck-anchored stochastic transport template is identified that provides a concrete experimental discriminator between deterministic tidal shear and irreducible directional fluctuations.

The results do not replace established field theories but show that several foundational structures of relativistic and quantum physics admit a unified statistical interpretation in terms of invariant null-directional kinematics.

Keywords: Null directional kinematics; spacetime emergence; directional statistics; information geometry; quantum decoherence; gauge connections

1 Introduction: Physics as Directional Statistics

The standard pillars of modern physics—Special Relativity (SR), General Relativity (GR), and Quantum Mechanics (QM)—are typically formulated from distinct primitives. Relativistic time dilation, gravitational acceleration, and quantum uncertainty are introduced through different foundational assumptions. Yet all known causal propagation ultimately occurs along null directions.

This observation motivates a statistical description in which the primitive local object is not a metric or wavefunction, but a normalized distribution of propagation directions on the null sphere.

In this view, many familiar structures of modern physics arise as statistical functionals of a directional ensemble:

- **Mean (first moment)** $\rightarrow$ transport/flow $\mathbf{v}$ (gravity-as-transport).
- **Symmetric overlap (Hellinger/Bhattacharyya affinity)** $\rightarrow$ Lorentz factor (time dilation).
- **Support duality** $\rightarrow$ Heisenberg uncertainty (Fourier).
- **Internal frame redundancy of directional amplitudes** $\rightarrow$ gauge connection (Yang–Mills structure).

In this perspective, the familiar geometric structures of relativistic physics—metric relations, causal structure, and gauge connections—appear not as primitive postulates but as effective descriptions of the statistical organization of null propagation. The main purpose is to state and justify this correspondence with minimal derivations, and to isolate a falsifiable prediction.

Furthermore, it will be demonstrated that the classical force fields of the standard paradigm, specifically gravitation and electromagnetism, need not be introduced as independent primitive entities. Rather, as shown in Section 7, both the Einstein field equations and Maxwell's equations are identified as the unique leading-order covariant closures of the emergent transport and internal frame redundancies once an effective local field description is assumed (locality, covariance, and second-order dynamics).

The remainder of the paper develops this program as follows. Section 2 introduces the null-directional ensemble as the primitive kinematic structure. Section 3 states the central statistical theorem and derives its SR, GR, QM, and gauge corollaries. Later sections introduce the directional strain functional and explore its experimental and cosmological consequences.

Novelty and Falsifiability. Many structural correspondences identified in this work (e.g., Lorentz dilation from symmetric overlap, gauge connections from internal redundancy) can be interpreted as reformulations of known geometric facts within a statistical language.

This reformulation is nevertheless nontrivial: it identifies a single directional statistical substrate from which several previously independent postulates of relativistic and quantum physics emerge as statistical functionals.

The genuinely new physical content of the framework lies in two areas:

(i) the finite-information lemma and resulting Planck-anchored stochastic transport template (Sec. 9), which predicts a momentum-quadratic decoherence floor linked to directional strain and admits direct experimental falsification;

(ii) the saturation limit $\chi \rightarrow 0$ of the directional overlap, which corresponds to complete polarization of the null ensemble and provides a kinematic criterion for gravitational collapse. In cosmological settings, this yields a horizon-scaled saturation mechanism capable of generating rare, direct-collapse supermassive black hole seeds in the early universe. The resulting high-mass tail behavior and redshift scaling of early black hole populations are developed quantitatively in [1].

2 Primitive Structure

Definition 2.1 (Null-directional ensemble)**.** *Let kinematics be described by an ensemble of future-directed null* four-*vectors* $k^\mu = (\omega/c, \mathbf{k})$ *satisfying the Minkowski null condition* $k^\mu k_\mu = 0$ *(signature* $-+++$*) and* $\omega > 0$*. Equivalently, the spatial part obeys*

$$|\mathbf{k}| = \frac{\omega}{c}. \tag{2.1}$$

For fixed frequency ω *(or after normalizing to unit direction), define the resolved propagation direction*

$$\hat{\mathbf{k}} \equiv \frac{\mathbf{k}}{|\mathbf{k}|} \in S^2. \tag{2.2}$$

Direction $\hat{\mathbf{k}}$ *is encoded by a normalized distribution*

$$P(\hat{\mathbf{k}}; \mathbf{x}, t) \geq 0, \qquad \int_{S^2} P(\hat{\mathbf{k}}; \mathbf{x}, t)\, d\Omega = 1. \tag{2.3}$$

Here $d\Omega$ *denotes the standard solid-angle measure on* S^2*. Observable quantities arise as statistical functionals of* P *[2].*

3 The Unification Theorem (Core)

The structural result is stated below, followed by compact corollaries for SR, GR, QM, and gauge structure.

Theorem 3.1 (Null-Directional Statistical Theorem)**.** *Assume* (2.1)–(2.3)*. Define the first moment (transport field)*

$$\mathbf{v}(\mathbf{x}, t) \equiv c \int_{S^2} \hat{\mathbf{k}}\, P(\hat{\mathbf{k}}; \mathbf{x}, t)\, d\Omega. \tag{3.1}$$

Then:

1. Special Relativity: *In 1D, once the local proper-time rate is identified with a symmetric directional-overlap functional of the null weights, the Lorentz factor is recovered exactly.*
2. Newtonian Gravity: *In stationary, irrotational flow, spatial transport of* $\mathbf{v}$ *yields a Newtonian potential* Φ *and acceleration* $\mathbf{g} = -\nabla\Phi$*.*
3. Quantum Mechanics: *If momentum corresponds to directional resolution and spatial localization restricts support, the Heisenberg inequality follows from Fourier duality.*
4. Gauge Structure: *If directional amplitudes carry an internal frame redundancy, covariance under local internal frame change introduces a connection transforming as a gauge field.*

Remark on Dynamics. While the primary focus of this section is the kinematic emergence of relativistic observables, the local $U(1)$ phase redundancy intrinsic to the spinor representation inherently mandates a geometric connection field. As will be established in Section 7, the Maxwell system is identified as the unique leading-order covariant closure for this connection once an effective local field description is assumed (locality, covariance, and second-order dynamics), just as the Einstein field equations are identified as the unique leading-order covariant closure for the local Lorentz redundancy of the null tetrad under the same assumptions.

3.1 Part A: Spacetime from moments (SR and weak-field limits)

Axis-reduced representation of the null-sphere state. The fundamental object of the framework is a normalized directional distribution $P(\hat{k}; x)$ defined on the local null sphere S^2_x, with

$$\int_{S^2} P(\hat{k}; x)\, d\Omega = 1, \qquad |\hat{k}| = 1.$$

Given a chosen local axis $\hat{n}$—for example the normalized dipole direction when $\langle \hat{k} \rangle \neq 0$, or an externally specified transport axis—the full directional state may be coarse-grained into forward and backward hemispherical weights

$$w_+(\hat{n}) = \int_{\hat{k}\cdot\hat{n}>0} P(\hat{k}; x)\, d\Omega, \qquad w_-(\hat{n}) = \int_{\hat{k}\cdot\hat{n}<0} P(\hat{k}; x)\, d\Omega, \tag{3.2}$$

which satisfy $w_+ + w_- = 1$. The following one-dimensional construction should therefore be understood as the minimal axis-reduced sector of the full null-sphere description, obtained by projecting the directional ensemble onto a chosen forward/backward decomposition. Under this reduction, parity reversal corresponds to $\hat{k} \mapsto -\hat{k}$, which exchanges the hemispherical weights $(w_+, w_-) \mapsto (w_-, w_+)$.

Special Relativity from directional overlap (minimal constitutive derivation). In one spatial dimension, let $w_\pm$ denote the weights assigned to the forward and backward null directions, with

$$w_+ + w_- = 1, \qquad v = c(w_+ - w_-). \tag{3.3}$$

Hence

$$w_\pm = \frac{1}{2}\left(1 \pm \frac{v}{c}\right). \tag{3.4}$$

The primitive null-directional ensemble fixes the kinematic weights $w_\pm$, but by itself does not yet determine the operational clock rate associated with a local state. A constitutive identification is therefore required.

Definition 3.2 (Constitutive clock functional)**.** *For a 1D reduced directional state $p = (w_+, w_-)$ and its parity-reversed state $q = (w_-, w_+)$, define the dimensionless local proper-time rate by the symmetric directional overlap*

$$\frac{d\tau}{dt} \;\equiv\; \chi(p, q) \;\equiv\; BC(p, q) = \sum_{i\in\{+,-\}} \sqrt{p_i q_i} = 2\sqrt{w_+ w_-}. \tag{3.5}$$

Equivalently,

$$\gamma \equiv \frac{dt}{d\tau} = \chi^{-1}. \tag{3.6}$$

This choice is not introduced as a theorem of the primitive ensemble alone, but as the canonical information-geometric constitutive law compatible with the required physical symmetries. In particular:

- **At rest:** $w_+ = w_- = 1/2$, so the state is identical to its reverse and the overlap is maximal:

  $$\chi = BC(p, q) = 1.$$

  The clock ticks at its maximal rate.

- **At complete directional polarization:** $(w_+, w_-) = (1, 0)$ or $(0, 1)$, so the state and its reverse have disjoint support and

$$\chi = BC(p, q) = 0.$$

  The overlap clock halts.

Among smooth symmetric overlap functionals on probability simplices, the Bhattacharyya coefficient is the natural minimal choice for the present framework: it is invariant under relabeling, multiplicative under independent composition, and depends only on the pointwise geometric mean of the directional weights [3]. These properties make it the canonical overlap functional for a statistical kinematics, even though other symmetric functionals may also be defined.

Remark (relation to full-sphere overlap). The 1D overlap $\chi = 2\sqrt{w_+ w_-}$ should be understood as the minimal axis-reduced form of the Bhattacharyya overlap on S^2, obtained by coarse-graining the full directional distributions $P(\hat{k})$ and $P(-\hat{k})$ into forward/backward hemispherical weights relative to $\hat{n}$.

Substituting the kinematic weights (3.4) into (3.5) gives

$$\chi = 2\sqrt{w_+ w_-} = 2\sqrt{\frac{1}{4}\left(1 + \frac{v}{c}\right)\left(1 - \frac{v}{c}\right)} = \sqrt{1 - \frac{v^2}{c^2}}. \tag{3.7}$$

Therefore

$$\gamma = \chi^{-1} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \tag{3.8}$$

Remark. Equation (3.8) is an exact consequence of the null-directional weights together with the constitutive identification (3.5). In this sense, Lorentz time dilation is recovered as the inverse directional self-overlap of the local state: increasing macroscopic polarization decreases symmetric directional support and therefore slows the overlap clock.

Saturation limit. The limit $\chi \to 0$ corresponds to complete directional polarization of the null ensemble. In the present framework this is interpreted as a kinematic saturation condition in which symmetric directional support vanishes. In macroscopic settings, such saturation defines a natural collapse threshold independent of classical curvature singularities.

3.1.1 Finite-throughput lemma: irreducible overlap fluctuations

The clock rate in 1D has been identified with the directional overlap

$$\chi = 2\sqrt{w_+ w_-}. \tag{3.9}$$

A distinct microstructural question is whether this overlap can be operationally realized as perfectly smooth, or whether finite substrate throughput necessarily induces fluctuations during a fixed coarse-graining interval [4, 5].

Proposition 3.3 (Finite-throughput inference bound)**.** *Assume that, within a local operational region and during a fixed coarse-graining interval, the 1D directional state is inferred from a finite effective number N_{eff} of statistically informative update-events.*

Let θ denote a one-parameter description of the directional bias (for example $\theta = w_+$, or equivalently $\theta = v/c$), and let $\hat{\theta}$ be an unbiased estimator constructed from these events. Then, under standard regularity conditions,

$$\mathrm{Var}(\hat{\theta}) \;\geq\; \frac{1}{\mathcal{I}_N(\theta)} \;\sim\; \frac{1}{N_{\mathrm{eff}}\,\mathcal{I}_1(\theta)}, \tag{3.10}$$

where $\mathcal{I}_N(\theta)$ is the Fisher information in the coarse-graining interval and $\mathcal{I}_1(\theta)$ is the Fisher information per effective event.

Consequently, for any smooth clock functional $\chi = \chi(\theta)$ and away from singular boundary points,

$$\mathrm{Var}(\hat{\chi}) \;\approx\; \left(\frac{d\chi}{d\theta}\right)^2 \mathrm{Var}(\hat{\theta}) \;\gtrsim\; \frac{1}{N_{\mathrm{eff}}}\left(\frac{d\chi}{d\theta}\right)^2 \frac{1}{\mathcal{I}_1(\theta)}. \tag{3.11}$$

Thus finite informational throughput implies a generic $1/N_{\mathrm{eff}}$ lower bound on the smoothness with which an overlap-based clock rate can be inferred.

Proof sketch. Equation (3.10) is the Cramér–Rao lower bound for unbiased estimation under standard regularity assumptions [6,7]. When information is extensive in the number of effective update-events, $\mathcal{I}_N(\theta) \sim N_{\mathrm{eff}}\mathcal{I}_1(\theta)$. Propagation of uncertainty through a smooth functional $\chi(\theta)$ gives (3.11). □

Physical realization assumption. Proposition 3.3 is an inference statement. To convert it into a physical statement about local clock behavior, an additional microphysical assumption is required:

During each operational coarse-graining interval, the local overlap clock is not evaluated from an underlying continuously given hidden parameter, but is physically realized by a finite stochastic aggregation of substrate update-events.

Under this realization assumption, the operationally realized clock rate itself inherits an irreducible finite-throughput fluctuation floor. The inference bound above then serves as the information-theoretic benchmark for the magnitude of those fluctuations.

Benchmark micro-model (independent binomial channels). A simple realization that exhibits the generic $1/N_{\mathrm{eff}}$ scaling is obtained by taking N independent directional update-events with outcomes in $\{+,-\}$ and probabilities (w_+, w_-). In this benchmark, $N_{\mathrm{eff}} = N$.

Corollary 3.4 (Interior-regime overlap fluctuations in the binomial benchmark)**.** *Let $n_+ \sim \mathrm{Binomial}(N, w_+)$,*

$$\hat{w}_+ = \frac{n_+}{N}, \qquad \hat{w}_- = 1 - \hat{w}_+, \qquad \hat{\chi} = 2\sqrt{\hat{w}_+\hat{w}_-}.$$

Assume

$$0 < w_+ < 1, \qquad N w_+ w_- \gg 1, \tag{3.12}$$

so that the distribution is away from the boundary and the first-order delta approximation is valid. Then

$$\mathrm{Var}(\hat{\chi}) = \frac{(w_+ - w_-)^2}{N} + O(N^{-2}) = \frac{v^2}{c^2\,N} + O(N^{-2}). \tag{3.13}$$

Proof sketch. For the binomial model,

$$\mathrm{Var}(\hat{w}_+) = \frac{w_+ w_-}{N}. \tag{3.14}$$

With

$$\chi(w_+) = 2\sqrt{w_+(1-w_+)}, \tag{3.15}$$

one has

$$\frac{d\chi}{dw_+} = \frac{w_- - w_+}{\sqrt{w_+ w_-}}. \tag{3.16}$$

Applying the delta method in the interior regime (3.12),

$$\mathrm{Var}(\hat{\chi}) \approx \left(\frac{w_- - w_+}{\sqrt{w_+ w_-}}\right)^2 \left(\frac{w_+ w_-}{N}\right) = \frac{(w_+ - w_-)^2}{N}. \tag{3.17}$$

Using $w_+ - w_- = v/c$ gives (3.13). □

Boundary behavior. Equation (3.13) is an interior asymptotic formula and should not be extrapolated uniformly to the special limits $w_+ = w_- = 1/2$ or $w_+ w_- \to 0$. At exact isotropy, the first derivative (3.16) vanishes, so the leading variance begins at second order, $\mathrm{Var}(\hat{\chi}) = O(N^{-2})$. Near complete polarization, the first-order delta approximation fails because (3.16) becomes singular; in the exactly polarized limit $(w_+, w_-) = (1, 0)$ or $(0, 1)$, one has $\hat{\chi} = 0$ almost surely and therefore $\mathrm{Var}(\hat{\chi}) = 0$.

Interpretation. Finite throughput therefore implies that the overlap clock is not generically perfectly smooth when it is realized through finitely many substrate events. In the independent-event benchmark and away from the boundary regimes, the fluctuation amplitude is quadratic in directional bias:

$$\mathrm{Var}(\hat{\chi}) \sim \frac{v^2}{c^2 N}.$$

If accumulated phase is proportional to the time integral of the realized local clock rate,

$$\phi(t) \propto \int^t \chi(t')\, dt',$$

then short-correlated fluctuations in χ induce phase diffusion with rate proportional to $\mathrm{Var}(\chi)$. In transport-defined regimes, where $p_{\mathrm{arm}} \propto m v_{\mathrm{arm}}$, the benchmark micro-model therefore yields the leading-order scaling

$$\Gamma_\phi \propto p_{\mathrm{arm}}^2$$

for the stochastic decoherence floor discussed in Sec. 9.

More generally, correlations among substrate update-events renormalize N into an effective number N_{eff} without changing the generic finite-throughput structure. Thus deterministic relativistic time dilation remains governed by the first moment of the directional state, while the discrete finite-throughput realization of that state generically underlays the smooth limit by stochastic clock jitter.

Scope of the stochastic prediction. The momentum-quadratic decoherence law,

$$\Gamma_\phi \propto p_{\mathrm{arm}}^2,$$

should be understood as the leading-order benchmark prediction of the finite stochastic realization introduced above, obtained explicitly in the independent-binomial interior regime. The more universal claim of the framework is weaker and more general: finite informational throughput implies a stochastic floor in the realized overlap clock rate that scales generically as $1/N_{\mathrm{eff}}$, with the precise velocity or momentum dependence determined by the underlying microphysical update statistics.

Ontic status in the stochastic realization. Within the stochastic realization assumption stated above, the fluctuations of $\hat{\chi}$ are not interpreted merely as epistemic uncertainty about a hidden smooth parameter. Rather, $\hat{\chi}$ is the operationally realized clock rate produced by a finite substrate update process during each coarse-graining interval. In that sense the finite-N_{eff} fluctuations are ontic: they correspond to genuine variability in local clock updates and hence to physical phase diffusion. The smooth geometric limit is recovered as $N_{\text{eff}} \to \infty$.

Newtonian limit as a transport consistency check. Let $\mathbf{v}(\mathbf{x}, t)$ be the first moment (3.1) and define the material acceleration

$$\mathbf{g} = \frac{D\mathbf{v}}{Dt} = \partial_t \mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{v}. \tag{3.18}$$

Using the identity

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \nabla\left(\frac{1}{2}v^2\right) - \mathbf{v} \times (\nabla \times \mathbf{v}), \tag{3.19}$$

and restricting to stationary, irrotational flow ($\partial_t \mathbf{v} = 0$, $\nabla \times \mathbf{v} = 0$),

$$\mathbf{g} = \nabla\left(\frac{1}{2}v^2\right) = -\nabla\Phi, \qquad \Phi \equiv -\frac{1}{2}v^2. \tag{3.20}$$

For a stationary spherically symmetric profile $v = v(r)$, one has

$$\mathbf{g} = \nabla\left(\frac{1}{2}v^2\right) = \frac{d}{dr}\left(\frac{1}{2}v(r)^2\right)\hat{\mathbf{r}}. \tag{3.21}$$

Calibrating the stationary radial transport profile to match the observed $1/r$ field (equivalently adopting the escape-speed scaling) via

$$v(r) = \sqrt{\frac{2GM}{r}} \tag{3.22}$$

gives $\frac{1}{2}v^2 = GM/r$ and therefore

$$\mathbf{g} = \nabla\left(\frac{GM}{r}\right) = -\frac{GM}{r^2}\hat{\mathbf{r}}, \qquad \Phi = -\frac{GM}{r}, \tag{3.23}$$

recovering the standard Newtonian potential (outside the source). Note that $\nabla(1/r) = -(1/r^2)\hat{\mathbf{r}}$ points inward, so with $\Phi = -GM/r$ one has $\mathbf{g} = -\nabla\Phi = -(GM/r^2)\hat{\mathbf{r}}$ as usual.

Corollary 3.5 (Newtonian limit from first-moment transport)**.** *A Newtonian-like potential flow arises from spatial transport of the first moment, with $\Phi = -\frac{1}{2}v^2$; Choosing the stationary transport profile*

$$v(r) = \sqrt{\frac{2GM}{r}},$$

which corresponds to the escape-speed scaling for radial transport, reproduces $\Phi = -GM/r$ outside a spherical source.

3.2 Part B: Quantum mechanics from support duality

Represent a pre-resolved directional state by an amplitude on the null sphere:

$$\Psi = \int_{S^2} \eta(\hat{\mathbf{k}}) \left|\hat{\mathbf{k}}\right\rangle \, d\Omega. \tag{3.24}$$

To connect directional amplitudes to the usual position–momentum Fourier pair, consider a narrowband (fixed $|\mathbf{k}| = k_0$) packet so that momentum is $\mathbf{p} \approx \hbar k_0 \hat{\mathbf{k}}$ and the direction $\hat{\mathbf{k}}$ labels momentum *orientation*. In the standard plane-wave expansion,

$$\psi(\mathbf{x}) \propto \int_{S^2} \eta(\hat{\mathbf{k}}) \, e^{ik_0 \hat{\mathbf{k}}\cdot\mathbf{x}} \, d\Omega, \tag{3.25}$$

this narrowband example makes the support-duality mechanism explicit in the angular (transverse-momentum) sector. Restricting spatial support of $\psi(\mathbf{x})$ forces broadening of the conjugate momentum distribution, here encoded by the angular spread of the directional amplitude η at fixed k_0. In the unrestricted case, restoring the full momentum-space amplitude $\tilde{\psi}(\mathbf{p})$ on $\mathbb{R}^3$ gives the standard Fourier pair $\psi(\mathbf{x}) \leftrightarrow \tilde{\psi}(\mathbf{p})$ and hence the usual Heisenberg inequality [8].

Interpreting momentum as directional resolution, spatial localization restricts spatial support and therefore broadens momentum support by Fourier duality, yielding

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}. \tag{3.26}$$

Corollary 3.6 (Uncertainty as incomplete directional resolution)**.** *Heisenberg uncertainty arises from Fourier duality between spatial localization and momentum-space concentration, interpreted in the present framework as a limit on directional resolution.*

4 Mass as Directional Information Constraint

The previous section identified gravity kinematically with spatial transport of the first moment $\mathbf{v}$. What remains is the identification of *source*: what in the null-ensemble ontology corresponds to mass-energy?

Since the primitive local object is the normalized directional distribution $P(\hat{\mathbf{k}}; x)$ on S^2, the only invariant scalar measuring departure from vacuum isotropy is informational. Deep conceptual connections between gravitational dynamics and thermodynamic or information-theoretic principles have been explored in several influential frameworks [9–12]. In the present approach, similar structural features arise from a different starting point: the directional statistics of null propagation. Gravitational sourcing is interpreted as the information cost required to maintain anisotropic directional structure in the underlying null ensemble.

4.1 Directional strain functional

Let the isotropic vacuum ensemble be

$$P_0(\hat{\mathbf{k}}) = \frac{1}{4\pi}. \tag{4.1}$$

Definition 4.1 (Directional strain)**.** *The local directional strain of the null ensemble is defined as the relative entropy*

$$\mathcal{I}(x) \equiv \int_{S^2} P(\hat{\mathbf{k}}; x) \ln\left(\frac{P(\hat{\mathbf{k}}; x)}{P_0(\hat{\mathbf{k}})}\right) \, d\Omega. \tag{4.2}$$

This quantity satisfies $\mathcal{I} \geq 0$ and vanishes if $P = P_0$. It therefore measures the minimal information cost required to maintain a structured directional ensemble relative to isotropic vacuum. Relative entropy plays a central role in information geometry and quantum field theory [13, 14].

4.2 Mass–strain postulate

Proposition 4.2 (Strain energy density)**.** *The* effective *energy density associated with the directional strain field is defined by the persistent informational constraint on the null ensemble:*

$$\rho_{\text{eff}}(x)c^2 = \Lambda_* \mathcal{I}(x), \tag{4.3}$$

where Λ_* sets the microscopic energy density scale of the substrate.

Bookkeeping of densities. Ordinary (localized) matter density $\rho_m(x)$ is distinguished from the distributed strain-sector energy density $\rho_{\text{eff}}(x)$ defined by Eq. (4.3). In the weak-field closure used below, ρ_m sources the strain field through a Poisson-type ansatz, while ρ_{eff} represents the effective stress-energy carried by the strain sector and may extend into regions where $\rho_m = 0$.

Remarks. (i) Λ_* has dimensions of energy density and may be written $\Lambda_* \sim \hbar c/\ell_*^3$ in terms of a microscopic length ℓ_*. (ii) Equation (4.3) is not an external GR assumption; it is the minimal identification of what "mass" means in a directional-statistics ontology: mass is the energetic cost of constraining null-directional degrees of freedom.

Physical interpretation of directional strain. In the null-ensemble picture, local transport capacity is determined by the symmetric support of available propagation directions. A reduction of accessible solid-angle measure corresponds to a loss of isotropic transport channels. Because clock rates are defined operationally by symmetric overlap of forward and backward null support, diminished angular support necessarily reduces proper-time accumulation. Spatial gradients in this transport deficit therefore produce differential clock rates and effective acceleration. In this sense, gravitational potential is interpreted as a macroscopic manifestation of directional information depletion.

4.3 Structural consequence: curvature as compatibility condition

Directional strain modifies the first moment $\mathbf{v}$ through transport. The material acceleration

$$\mathbf{g} = \frac{D\mathbf{v}}{Dt} \tag{4.4}$$

must therefore be compatible with spatial gradients of $\mathcal{I}$. To leading order in weak strain,

$$\nabla \cdot \mathbf{g} \;\propto\; \nabla^2 \mathcal{I}. \tag{4.5}$$

If the strain field is locally sourced by ordinary matter density ρ_m, this yields the schematic sourcing relation

$$\nabla \cdot \mathbf{g} \;\propto\; \rho_m, \tag{4.6}$$

with the explicit weak-field closure given below by the Poisson-type ansatz $\nabla^2 \mathcal{I} \propto -\rho_m$ (Eq. (5.18)). This recovers the structural form of Poisson's equation without postulating Newtonian gravity as primitive.

Corollary 4.3 (Field equation as strain compatibility)**.** *Gravitational field equations arise as macroscopic compatibility conditions ensuring that directional strain $\mathcal{I}$ and transport $\mathbf{v}$ evolve consistently.*

Interpretation. In this framework, mass does not "curve spacetime" as an independent axiom. Rather:

- Mass is persistent information constraint on null-directional degrees of freedom.
- Transport of the first moment encodes effective geometry.
- Field equations express compatibility between informational strain and null transport.

The relativistic generalization follows by replacing $\nabla \cdot \mathbf{g}$ with curvature scalars constructed from the effective transport geometry, yielding a covariant strain-balance equation.

5 Covariant Closure: Einstein Equation as Strain-Balance

The preceding sections define: (i) a kinematic transport field $\mathbf{v}$ as the first moment of the null ensemble, and (ii) a scalar directional strain functional $\mathcal{I}(x)$ as the relative entropy of $P(\hat{\mathbf{k}}; x)$ with respect to isotropy. Obtaining gravitational *field equations* requires specifying how the emergent geometry responds to strain.

The goal here is not to assume GR as a primitive, but to show that once one accepts an *emergent metric description* of transport geometry, the field equation is essentially *forced* by covariance and consistency.

5.1 Minimal axioms for a metric effective description

Assume the null-ensemble admits an effective macroscopic geometric description by a Lorentzian metric $g_{\mu\nu}(x)$ in the regime of interest. This assumption is an effective macroscopic hypothesis rather than a microscopic derivation; the result of this section is that, once such a metric description is admitted, its dynamical closure is uniquely constrained. The following conditions are the minimal structural requirements for a local field theory of that metric:

(A1) *Locality:* the field equation is local in x.

(A2) *General covariance:* the field equation is tensorial under diffeomorphisms.

(A3) *Second-order dynamics:* the metric field equation contains at most second derivatives of $g_{\mu\nu}$ (avoiding higher-derivative instabilities in the minimal closure).

(A4) *Conservation:* the source tensor obeys $\nabla_\mu T^{\mu\nu} = 0$, consistent with diffeomorphism invariance of the underlying null-ensemble description.

Logical Status of the Closure. The null-directional framework does not derive the full nonlinear Einstein equations from microscopic first principles in the present work. Rather, it establishes that:

(i) the null ensemble defines a local transport geometry and scalar strain field,

(ii) strain provides a covariant candidate source sector,

(iii) once an effective metric description is assumed,

the macroscopic gravitational field equation is uniquely fixed (up to a cosmological term) by locality, general covariance, second-order dynamics, and covariant conservation in four dimensions (Lovelock's theorem).

Thus the Einstein equations appear not as an independent postulate, but as the unique consistent covariant closure of the emergent transport geometry.

5.2 Strain as matter: stress tensor from the null ensemble

To turn the scalar strain $\mathcal{I}$ into a covariant source, it is treated as a scalar functional of the local null ensemble in an orthonormal frame. Formally, the distribution $P(\hat{\mathbf{k}}; x)$ is defined on the local celestial sphere of the tangent space; its relative entropy to isotropy defines $\mathcal{I}(x)$ as a scalar.

The minimal covariant "strain energy" action is now introduced as

$$S_{\text{strain}} \equiv -\int d^4x \sqrt{-g}\, \Lambda_* \mathcal{I}(x), \tag{5.1}$$

where Λ_* sets the microscopic energy density scale (cf. Eq. (4.3)).

Definition 5.1 (Strain stress-energy tensor)**.** *The effective stress-energy tensor associated with directional strain is defined by*

$$T^{\mu\nu}_{\text{strain}} \equiv \frac{2}{\sqrt{-g}} \frac{\delta S_{\text{strain}}}{\delta g_{\mu\nu}} = -2\Lambda_* \frac{\delta \mathcal{I}}{\delta g_{\mu\nu}} - \Lambda_* \mathcal{I}\, g^{\mu\nu}. \tag{5.2}$$

Remark (metric dependence and conservation). In general, $\mathcal{I}(x)$ depends on the metric through the choice of local orthonormal frame and measure on the celestial sphere (equivalently through the tetrad $e^a{}_\mu$), so the full variational definition (5.2) must be retained to ensure covariant conservation. Indeed, diffeomorphism invariance of the total action implies $\nabla_\mu T^{\mu\nu}_{\text{strain}} = 0$ when the $\mathcal{I}$-sector equations of motion (or constitutive closure) are satisfied.

For intuition it is convenient to separate an isotropic "potential-like" piece,

$$T^{\mu\nu}_{\text{strain}} \supset -\Lambda_* \mathcal{I}\, g^{\mu\nu},$$

from additional derivative/anisotropic contributions contained in

$$-2\Lambda_* \frac{\delta \mathcal{I}}{\delta g_{\mu\nu}}.$$

In weak-strain, slowly varying regimes these derivative terms may be subleading in magnitude in some components, but they are generically required for exact conservation when $\nabla_\mu \mathcal{I} \neq 0$. In the weak-field calibration below, $\mathcal{I}$ is treated as an independent effective scalar degree of freedom obeying a phenomenological Poisson closure, leaving its fully covariant dependence $\mathcal{I}[g]$ implicit. Additional non-strain matter contributions may be included in the usual way by adding S_{matter} and defining $T^{\mu\nu}_{\text{matter}}$ similarly.

5.3 Einstein equation as the unique covariant strain-balance law

Theorem 5.2 (GR-compatible covariant strain balance)**.** *Under axioms (A1)–(A4), and assuming the effective macroscopic metric description is defined on a four-dimensional Lorentzian manifold, any local covariant second-order metric closure consistent with covariant conservation is constrained by Lovelock's theorem to take the form*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \tag{5.3}$$

where $T_{\mu\nu}$ is the total conserved stress-energy tensor. If the only source is directional strain, $T_{\mu\nu} = T^{\text{strain}}_{\mu\nu}$ from Definition 5.1. The constants (κ, Λ) are fixed by calibration to the weak-field limit and cosmological boundary conditions.

Calibration to the weak-field limit. Expanding $g_{00} \approx -(1 + 2\Phi/c^2)$ and retaining leading order, the 00 component of Eq. (5.3) yields a Poisson-type equation

$$\nabla^2\Phi \;=\; 4\pi G\,\rho_{\text{eff}}, \tag{5.4}$$

with $\kappa = 8\pi G/c^4$. In the present framework, ρ_{eff} is determined by the identification $\rho_{\text{eff}}c^2 = \Lambda_*\mathcal{I}$ (Eq. (4.3)) together with the constitutive dependence of $\mathcal{I}$ on the local orthonormal frame (and hence on $g_{\mu\nu}$) in the effective description.

5.4 Constraint-manifold micro-model (missing solid angle)

A minimal microscopic realization of "mass as directional constraint" is now supplied that is native to the null-ensemble ontology and yields closed-form strain $\mathcal{I}$.

Model. Let the vacuum distribution be isotropic on S^2: $P_0(\hat{\mathbf{k}}) = 1/(4\pi)$. Introduce matter as a local *constraint manifold* that suppresses a subset of directions. Concretely, at each spacetime point x let $\mathcal{C}(x) \subset S^2$ be the *allowed* set of directions with solid angle

$$\Omega(x) \equiv \int_{\mathcal{C}(x)} d\Omega, \qquad 0 < \Omega(x) \le 4\pi. \tag{5.5}$$

The constrained ensemble is taken to be maximally uninformative *given the constraint*: uniform on $\mathcal{C}(x)$ and zero elsewhere,

$$P(\hat{\mathbf{k}}; x) = \begin{cases} \Omega(x)^{-1}, & \hat{\mathbf{k}} \in \mathcal{C}(x), \\ 0, & \hat{\mathbf{k}} \notin \mathcal{C}(x). \end{cases} \tag{5.6}$$

This is the unique maximum-entropy distribution compatible with the restriction that "only the directions in $\mathcal{C}$ are accessible." It is the least biased distribution possible; any non-uniform assignment on $\mathcal{C}(x)$ would encode additional, unjustified microscopic information beyond the mere physical presence of the geometric constraint. Therefore, this represents the minimal strain model. The constraint manifold should be interpreted as an effective coarse-grained description of how matter states bias the local directional ensemble, rather than as a literal mechanical obstruction of light rays.

Closed-form strain. The directional strain (relative entropy to isotropy) becomes

$$\begin{aligned} \mathcal{I}(x) &= \int_{S^2} P(\hat{\mathbf{k}}; x)\, \ln\!\left(\frac{P(\hat{\mathbf{k}}; x)}{P_0(\hat{\mathbf{k}})}\right) d\Omega \\ &= \int_{\mathcal{C}(x)} \frac{1}{\Omega(x)} \ln\!\left(\frac{(1/\Omega(x))}{(1/4\pi)}\right) d\Omega \\ &= \ln\!\left(\frac{4\pi}{\Omega(x)}\right). \end{aligned} \tag{5.7}$$

This closed form is specific to the uniform-on-$\mathcal{C}$ micro-model; for a general distribution P, $\mathcal{I}$ is defined by the full relative entropy (4.2), and Ω should be understood as an effective constrained support proxy. Thus, "mass" corresponds to a local reduction of accessible directional solid angle.

5.4.1 Covariant definition and transport of the constraint manifold

To couple the constraint-manifold model to an emergent Lorentzian geometry, the allowed directional set $\mathcal{C}(x) \subset S^2$ is defined in a local orthonormal frame, and its transport law is specified along matter worldlines. This supplies a covariant evolution equation for $\Omega(x)$ and closes the conservation structure.

Local celestial sphere in the tangent space. Let $e^\mu_a(x)$ be an orthonormal tetrad, $g_{\mu\nu} e^\mu_a e^\nu_b = \eta_{ab}$, with $\eta_{ab} = \mathrm{diag}(-1, 1, 1, 1)$. In this frame, future-directed null directions are parameterized by

$$\ell^a(\hat{\mathbf{k}}) = (1, \hat{\mathbf{k}}), \qquad \hat{\mathbf{k}} \in S^2, \qquad \eta_{ab}\ell^a\ell^b = 0. \tag{5.8}$$

The corresponding spacetime null vector is $\ell^\mu(\hat{\mathbf{k}}) = e^\mu_a \ell^a$. Thus the directional distribution $P(\hat{\mathbf{k}}; x)$ is naturally a scalar density on the local celestial sphere of the tangent space.

Constraint set as a frame-defined indicator. The constraint manifold is represented by an indicator function

$$\Xi(\hat{\mathbf{k}}; x) = \begin{cases} 1, & \hat{\mathbf{k}} \in \mathcal{C}(x), \\ 0, & \hat{\mathbf{k}} \notin \mathcal{C}(x), \end{cases} \tag{5.9}$$

so that the accessible solid angle is

$$\Omega(x) = \int_{S^2} \Xi(\hat{\mathbf{k}}; x)\, d\Omega. \tag{5.10}$$

In what follows, the notation is slightly overloaded: $\Omega(x)$ represents the geometric solid-angle fraction (5.10) multiplied by an implicit channel-density factor per comoving operational cell. This density factor is unity in the static calibration but need not be conserved under expansion, motivating the continuity equation (5.13). The maximally uninformative constrained ensemble is then

$$P(\hat{\mathbf{k}}; x) = \frac{\Xi(\hat{\mathbf{k}}; x)}{\Omega(x)}. \tag{5.11}$$

Matter-attached transport law (advection). Let $u^\mu(x)$ be the timelike 4-velocity field of the matter configuration sourcing the constraint. The minimal covariant assumption is that the geometric *shape* of the constraint on the local celestial sphere is an intrinsic property of the matter state, meaning Ξ is advected along u^μ:

$$u^\mu \nabla_\mu \Xi(\hat{\mathbf{k}}; x) = 0. \tag{5.12}$$

Channel-number conservation and the Ω continuity equation. While the geometric footprint Ξ is advected, the actual number of available physical channels dilutes if the comoving volume expands. Therefore, define the effective channel-number density $n_\Omega(x) \equiv \Omega(x)$, where Ω acts as a proxy for the *channel count per comoving operational cell*. A minimal constitutive conservation postulate is that the total number of accessible channels in a comoving 3-volume is conserved:

$$\nabla_\mu\big(\Omega\, u^\mu\big) = 0. \tag{5.13}$$

Equivalently, in comoving form,

$$u^\mu \nabla_\mu \ln \Omega = -\nabla_\mu u^\mu \equiv -\theta, \tag{5.14}$$

where θ is the expansion scalar of the matter congruence. Equation (5.14) is the covariant form of the channel-conservation rule $\Omega V = \text{const}$ used in the micro-model.

Implication for strain and mass-density evolution. With $\mathcal{I}(x) = \ln\left(\frac{4\pi}{\Omega(x)}\right)$, Eq. (5.14) implies

$$u^\mu \nabla_\mu \mathcal{I} = \theta. \tag{5.15}$$

The strain identification $\rho_{\text{eff}} c^2 = \Lambda_* \mathcal{I}$ yields a covariant evolution equation for the effective strain density ρ_{eff}:

$$u^\mu \nabla_\mu \rho_{\text{eff}} = \frac{\Lambda_*}{c^2}\,\theta. \tag{5.16}$$

This makes explicit that, in the constraint-manifold picture, gravitational source strength is tied to the kinematics of the comoving flow through the expansion θ of matter.

5.5 Static spherical source: strain field, $\Omega(r)$ profile, and weak-field metric

A localized mass must generate a gravitational field in exterior vacuum. In the present ontology this means: although ordinary matter density $\rho_m(x)$ is localized, the directional strain field $\mathcal{I}(x)$ (and hence $\Omega(x)$) must generally extend into vacuum. This subsection provides a minimal static, spherically symmetric closure showing how a point mass induces a $1/r$ exterior field and fixes the $\Omega(r)$ profile.

Scope of the spherical closure. The static spherical construction in this subsection is a *weak-field* calibration designed to reproduce the exterior $1/r$ potential and leading-order redshift. It is not claimed to reproduce the exact Schwarzschild geometry at strong field; extending the closure to fully nonlinear equivalence (or identifying the correct strong-field constitutive law for $\mathcal{I}$ and Ω) is deferred to future work.

5.5.1 Strain field as the long-range mediator

Definition 5.3 (Strain field)**.** *The scalar strain field $\mathcal{I}(x)$ is defined by the closed-form constraint-manifold relation*

$$\mathcal{I}(x) = \ln\left(\frac{4\pi}{\Omega(x)}\right), \tag{5.17}$$

so that $\Omega = 4\pi e^{-\mathcal{I}}$.

The key physical input required for a long-range gravitational field is that ordinary matter acts as a *source* for $\mathcal{I}$. The arguments in Sec. 4 fix only the *structural* relation that a divergence of the transport acceleration must track spatial variation of strain (hence a Poisson-like form), but they do not by themselves determine the *dynamical* constitutive law for $\mathcal{I}$ nor its calibrated coefficient; Eq. (5.18) is therefore introduced as an explicit weak-field closure rather than derived from Newtonian gravity.

Minimal dynamical ansatz. In the static weak-field regime, the strain field $\mathcal{I}(x)$ must (i) propagate information about localized directional constraints, (ii) respect locality, and (iii) reduce to a linear superposition principle for widely separated sources.

The simplest scalar field equation satisfying these requirements in Euclidean three-space is the Poisson equation. Equation (5.18) should therefore be understood as the minimal weak-field dynamical ansatz consistent with locality and superposition, rather than as a fundamental postulate derived from a complete microscopic theory of the null ensemble.

In this subsection, the standard Newtonian weak-field gauge on a fixed spatial slice is adopted, so ∇^2 denotes the flat-space Laplacian to leading order. The minimal weak-field, static closure is taken to be a Poisson-type equation

$$\nabla^2 \mathcal{I}(\mathbf{x}) = -\,\alpha\,\rho_m(\mathbf{x}), \tag{5.18}$$

where α is a coupling constant to be fixed by calibration to Newton's law. Here $\rho_m(\mathbf{x})$ denotes the localized ordinary matter density (not the distributed strain-sector density ρ_{eff}); the equation is a weak-field phenomenological closure specifying how matter sources the strain field $\mathcal{I}$.

This is the minimal local linear scalar field equation compatible with: (a) spherical symmetry, (b) weak-field superposition, and (c) recovery of the observed $1/r$ exterior potential in the static limit. More general nonlinear or higher-derivative closures are possible, but Eq. (5.18) represents the simplest weak-field assumption consistent with empirical gravity.

Status of Eq. (5.18)**.** Equation (5.18) is adopted as a minimal *phenomenological* closure in the static weak-field regime. It enforces locality and linear superposition while matching the observed Newtonian limit through calibration of α. A derivation of the $\mathcal{I}$-dynamics from an underlying null-ensemble microdynamics (or from a variational principle for $\mathcal{I}$) remains an open step of the program. Equation (5.18) should therefore be interpreted as a phenomenological weak-field closure rather than a first-principles consequence of the ensemble dynamics.

Equation (5.18) therefore encodes, at leading order, the precise mathematical statement that mass produces strain on the null substrate. The linearity of Eq. (5.18) is strictly a weak-field approximation; nonlinear completions consistent with the full covariant closure are expected at higher strain.

5.5.2 Exterior solution for a point mass

For a point mass M at the origin, $\rho_m(\mathbf{x}) = M\,\delta^{(3)}(\mathbf{x})$, the exterior region ($r > 0$) satisfies $\nabla^2\mathcal{I} = 0$ with the spherically symmetric solution

$$\mathcal{I}(r) = \frac{C}{r}, \qquad r > 0, \tag{5.19}$$

with C determined by integrating Eq. (5.18) over a small sphere:

$$\int \nabla^2\mathcal{I}\; d^3x = \oint \nabla\mathcal{I}\cdot d\mathbf{A} = -\,\alpha M. \tag{5.20}$$

Using $\mathcal{I} = C/r$ gives $\nabla\mathcal{I} = -(C/r^2)\hat{\mathbf{r}}$ and thus $\oint \nabla\mathcal{I}\cdot d\mathbf{A} = -4\pi C$, so

$$C = \frac{\alpha M}{4\pi}. \tag{5.21}$$

Therefore

$$\boxed{\mathcal{I}(r) = \frac{\alpha M}{4\pi r} \qquad (r > 0).} \tag{5.22}$$

5.6 Null focusing closure: curvature from missing solid angle

The null-ensemble ontology privileges congruences of null rays as primitive. In GR, the geometric content relevant to light propagation is encoded in the optical scalars of a null congruence; in particular, curvature enters through the Raychaudhuri focusing term $R_{\mu\nu}k^\mu k^\nu$ [15]. The constraint-manifold variable $\Omega(x)$ provides a natural substrate correction to optical expansion, yielding a covariant "strain–curvature" relation and fixing the spatial curvature once the clock-rate map has fixed g_{00}.

5.6.1 Optical area vs. effective (substrate) optical area

Let k^μ be the tangent to an affinely-parameterized null geodesic congruence, $k^\nu \nabla_\nu k^\mu = 0$, with affine parameter λ. Let $A(\lambda)$ be the geometric cross-sectional area of a small pencil of rays orthogonal to k^μ. The standard expansion scalar is

$$\theta \;\equiv\; \nabla_\mu k^\mu \;=\; \frac{d}{d\lambda} \ln A. \tag{5.23}$$

In the constraint-manifold model, the substrate does not necessarily support the full 4π solid angle of null directions. The *effective optical area* carried by the null substrate is defined as

$$A_{\text{eff}}(\lambda) \;\equiv\; A(\lambda)\, \frac{\Omega(x(\lambda))}{4\pi} = A(\lambda)\, e^{-\mathcal{I}(x(\lambda))}. \tag{5.24}$$

The corresponding *effective expansion* is

$$\theta_{\text{eff}} \;\equiv\; \frac{d}{d\lambda} \ln A_{\text{eff}} = \theta + \frac{d}{d\lambda} \ln\!\left(\frac{\Omega}{4\pi}\right) = \theta - \frac{d\mathcal{I}}{d\lambda}. \tag{5.25}$$

Interpretation. θ measures geometric spreading in the emergent metric, while θ_{eff} measures spreading of *available null channels* in the substrate. In vacuum isotropy ($\Omega = 4\pi$, $\mathcal{I} = 0$), $\theta_{\text{eff}} = \theta$.

5.6.2 Raychaudhuri equation and the strain contribution

For a hypersurface-orthogonal null congruence (vanishing twist) the null Raychaudhuri equation is

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - \; R_{\mu\nu} k^\mu k^\nu, \tag{5.26}$$

where $\sigma_{\mu\nu}$ is the shear tensor. For the idealized static, spherically symmetric calibration considered here, attention is restricted to the radial congruence for which shear may consistently be taken to vanish ($\sigma_{\mu\nu} = 0$) at leading order (see e.g. [16–18]).

Using $\theta = \theta_{\text{eff}} + d\mathcal{I}/d\lambda$ from Eq. (5.25), the focusing term can be rewritten as

$$\begin{aligned} R_{\mu\nu} k^\mu k^\nu &= -\frac{d\theta}{d\lambda} - \frac{1}{2}\theta^2 \\ &= -\frac{d}{d\lambda}\left(\theta_{\text{eff}} + \frac{d\mathcal{I}}{d\lambda}\right) - \frac{1}{2}\left(\theta_{\text{eff}} + \frac{d\mathcal{I}}{d\lambda}\right)^2. \end{aligned} \tag{5.27}$$

In regimes where the substrate expansion θ_{eff} is slowly varying compared with the strain-gradient term (e.g. weak fields, small pencils), a leading-order heuristic closure for the strain contribution is adopted. Specifically, in the exterior weak-field solution where $\mathcal{I}(r) \propto 1/r$, the retained terms scale as $\dot{\mathcal{I}} \sim r^{-2}\dot{r}$; the approximation therefore corresponds to neglecting $\dot{\theta}_{\text{eff}}$ and θ_{eff}^2 relative to $\ddot{\mathcal{I}}$ and $\dot{\mathcal{I}}^2$ along the congruence:

$$R_{\mu\nu} k^\mu k^\nu \;\approx\; -\frac{d^2\mathcal{I}}{d\lambda^2} - \frac{1}{2}\left(\frac{d\mathcal{I}}{d\lambda}\right)^2 \qquad \text{(spherical radial, } \sigma = 0\text{)}. \tag{5.28}$$

Equation (5.28) is the ontology-native statement that "mass produces strain on the null substrate" in a directly null-geometric form: the Ricci focusing encountered by null rays is controlled by derivatives of the directional strain along the congruence.

5.6.3 From null-projected curvature to Einstein equations

Define the effective stress tensor of the strain sector by requiring its null projection to match the strain focusing:

$$\boxed{\kappa\, T_{\mu\nu}^{(\mathrm{strain})} k^\mu k^\nu \;\equiv\; -\frac{d^2\mathcal{I}}{d\lambda^2} - \frac{1}{2}\left(\frac{d\mathcal{I}}{d\lambda}\right)^2 + \cdots,} \tag{5.29}$$

where the ellipsis denotes subleading terms arising from θ_{eff} variations and/or shear in less symmetric situations. Here $d/d\lambda \equiv k^\mu \nabla_\mu$ for an affinely parameterized congruence, so the leading terms may be read covariantly as $k^\mu k^\nu \nabla_\mu \nabla_\nu \mathcal{I}$ and $(k^\mu \nabla_\mu \mathcal{I})^2$ (up to sign conventions). The tensor $T_{\mu\nu}^{(\mathrm{strain})}$ is defined implicitly through its null projections; the existence of a local symmetric tensor realizing these projections follows from standard polarization identities in Lorentzian signature (assuming $T_{\mu\nu}$ is smooth and symmetric, and the null-projection equality holds in an open cone of directions at each point).

If the field equation is required to hold for all null directions k^μ at a point (i.e. for all congruence pencils), then Eq. (5.29) determines the strain sector through its null projections. Equivalently, if two symmetric tensors $X_{\mu\nu}$ and $Y_{\mu\nu}$ satisfy

$$X_{\mu\nu} k^\mu k^\nu = Y_{\mu\nu} k^\mu k^\nu$$

for all null k^μ at a point, then their difference must be proportional to the metric,

$$X_{\mu\nu} - Y_{\mu\nu} = \beta\, g_{\mu\nu},$$

for some scalar β. Thus the null-projection data fixes the tensor uniquely up to the usual trace (cosmological-term) ambiguity, yielding the covariant closure

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \left(T_{\mu\nu}^{(\mathrm{strain})} + T_{\mu\nu}^{(\mathrm{m})} \right), \tag{5.30}$$

where $T_{\mu\nu}^{(\mathrm{m})}$ denotes ordinary matter (localized source) and $T_{\mu\nu}^{(\mathrm{strain})}$ is the distributed substrate strain sector determined by $\mathcal{I}$.

Status of the result. The above construction does not claim to replace general relativity. Rather, it supplies a microphysical interpretation of gravitational source in terms of directional strain while remaining compatible with GR as the unique local covariant metric closure in four dimensions. The novelty lies in the identification of $\mathcal{I}(x)$ as the primitive quantity whose gradients govern both clock rate and null focusing.

5.6.4 Mapping strain to clock rate and g_{00}

To connect $\mathcal{I}$ to the effective metric, an operational relation is required between directional accessibility and local clock rate. The constraint-manifold model provides a natural candidate: if accessible null channels are reduced from 4π to Ω, then the symmetric "overlap capacity" of the ensemble is reduced by the factor $\sqrt{\Omega/4\pi}$. The minimal mapping is adopted

$$\frac{d\tau}{dt} \equiv \sqrt{\frac{\Omega(x)}{4\pi}} = e^{-\mathcal{I}(x)/2}. \tag{5.31}$$

In a static metric, $d\tau/dt = \sqrt{-g_{00}}$, hence

$$\boxed{g_{00}(x) = -\, e^{-\mathcal{I}(x)}.} \tag{5.32}$$

Note that this exponential map is chosen as a minimal, ontology-native clock-rate closure. The exponential form is the minimal positive monotone map that translates additive informational strain into multiplicative clock-rate reduction, inherently guaranteeing that the metric signature remains preserved ($g_{00} < 0$). While it reproduces the correct Newtonian limit once $\mathcal{I}(r) = 2GM/(c^2 r)$, it need not coincide with the exact Schwarzschild form beyond leading order.

In the weak-field limit, write $g_{00} \approx -(1 + 2\Phi/c^2)$. Expanding Eq. (5.32) for small $\mathcal{I}$ gives

$$g_{00} = -e^{-\mathcal{I}} \approx -(1 - \mathcal{I}), \qquad \Rightarrow \qquad \Phi \approx -\frac{c^2}{2}\,\mathcal{I}. \tag{5.33}$$

5.6.5 Fixing the spatial curvature g_{rr} from null focusing

In static spherical symmetry, write the metric in curvature coordinates as

$$ds^2 = -e^{2\Phi(r)} c^2 dt^2 + e^{2\Lambda(r)} dr^2 + r^2 d\Omega^2. \tag{5.34}$$

The clock-rate mapping of the constraint-manifold model fixes g_{00} via Eq. (5.32), i.e. $e^{2\Phi(r)} = e^{-\mathcal{I}(r)}$. The remaining function $e^{2\Lambda(r)}$ is fixed by requiring that the radial null congruence exhibits the focusing implied by the substrate effective area $A_{\text{eff}} = r^2\Omega(r)$.

Concretely, for a radial null geodesic $k^\mu = (\dot{t}, \dot{r}, 0, 0)$, the geometric area of a small pencil is $A = 4\pi r^2$ while the substrate area is $A_{\text{eff}} = r^2\Omega(r)$. Using Eq. (5.25) with $\theta = d(\ln r^2)/d\lambda = 2\dot{r}/r$ and $\mathcal{I} = \ln(4\pi/\Omega)$ gives

$$\theta_{\text{eff}} = \frac{2\dot{r}}{r} - \dot{\mathcal{I}}. \tag{5.35}$$

The Raychaudhuri equation (5.26) then relates $\dot{r}$ and its derivative (along λ) to $R_{\mu\nu}k^\mu k^\nu$, which is in turn fixed by $\mathcal{I}(r)$ through Eq. (5.28). Since $\dot{r}$ depends on $e^{2\Lambda(r)}$ via the null condition $ds^2 = 0$, this constrains $e^{2\Lambda(r)}$ once a constitutive closure for θ_{eff} is specified; in the weak-field regime, this consistently reproduces the standard leading-order Schwarzschild spatial scaling.

Outcome. Thus, in the null-ensemble ontology, the spatial curvature function g_{rr} is not an independent postulate: it is constrained (up to boundary conditions and closure choice) by the consistency of null focusing with the missing-solid-angle profile $\Omega(r)$ (equivalently the strain field $\mathcal{I}(r)$) that already fixed the clock rate.

5.6.6 Fixing the coupling by Newton's law

Insert Eq. (5.33) into the standard Poisson equation $\nabla^2\Phi = 4\pi G\rho_m$:

$$\nabla^2\left(-\frac{c^2}{2}\mathcal{I}\right) = 4\pi G\rho_m \quad \Rightarrow \quad \nabla^2\mathcal{I} = -\frac{8\pi G}{c^2}\rho_m. \tag{5.36}$$

Thus the strain-source coupling in Eq. (5.18) is fixed to be

$$\boxed{\alpha = \frac{8\pi G}{c^2}.} \tag{5.37}$$

With this choice, the point-mass exterior strain becomes

$$\boxed{\mathcal{I}(r) = \frac{2GM}{c^2 r}.} \tag{5.38}$$

Then Eq. (5.33) yields $\Phi(r) = -GM/r$ as required.

5.6.7 Predicted $\Omega(r)$ profile

Using $\Omega = 4\pi e^{-\mathcal{I}}$ with Eq. (5.38) gives

$$\boxed{\Omega(r) = 4\pi \, \exp\left(-\frac{2GM}{c^2 r}\right), \qquad r > 0.} \tag{5.39}$$

Distinctive ontological content. While the weak-field metric agrees with Schwarzschild at leading order, the variable $\Omega(r)$ has no direct analogue in classical GR. It encodes the microscopic directional-channel deficit underlying the macroscopic geometry. At present it does not introduce additional classical observables beyond those captured by the metric, but it provides a candidate microscopic variable whose fluctuations can generate experimentally testable decoherence signatures as discussed in Sec. 9.

Thus the null substrate's accessible directional solid angle decreases smoothly near a mass and approaches 4π at infinity. Importantly, this provides an explicit micro-to-macro bridge: the exterior gravitational field corresponds to an exterior strain field and a nontrivial $\Omega(r)$ profile even where $\rho_m = 0$.

Weak-field consistency. For $r \gg 2GM/c^2$,

$$\Omega(r) = 4\pi \left[1 - \frac{2GM}{c^2 r} + \mathcal{O}\left(\frac{GM}{c^2 r}\right)^2\right], \tag{5.40}$$

so the fractional directional deficit is $\varepsilon(r) \equiv 1 - \Omega/4\pi \approx 2GM/(c^2 r)$.

Interpretation. Equations (5.38)–(5.39) implement the statement "mass produces strain on the null substrate" in a way that reproduces the observed weak-field potential and yields a definite prediction for how the directional constraint variable Ω behaves in the exterior region.

Conservation of stress-energy (closure condition). Given a constitutive relation $p = p(\Omega)$ (equivalently $p = p(\rho_{\text{eff}})$) and the continuity law (5.13), the perfect-fluid conservation equation $\nabla_\mu T^{\mu\nu} = 0$ reduces to the standard energy balance along u^μ:

$$u_\nu \nabla_\mu T^{\mu\nu} = 0 \quad \Longleftrightarrow \quad u^\mu \nabla_\mu (\rho_{\text{eff}} c^2) + (\rho_{\text{eff}} c^2 + p)\, \theta = 0. \tag{5.41}$$

Thus, once a micro-motivated $p(\Omega)$ is specified, Eq. (5.13) provides a covariant route to $\nabla_\mu T^{\mu\nu} = 0$ without imposing it as an independent axiom.

Mass density. Using the mass–strain identification (Proposition 4.2),

$$\rho_{\text{eff}}(x) c^2 = \Lambda_* \, \mathcal{I}(x) = \Lambda_* \, \ln\left(\frac{4\pi}{\Omega(x)}\right). \tag{5.42}$$

For weak constraint (small deficit), write $\Omega(x) = 4\pi\,[1 - \varepsilon(x)]$ with $0 < \varepsilon \ll 1$, giving

$$\mathcal{I}(x) = \ln \frac{1}{1 - \varepsilon(x)} = \varepsilon(x) + \mathcal{O}(\varepsilon^2), \qquad \rho_{\text{eff}}(x) c^2 = \Lambda_* \, \varepsilon(x) + \mathcal{O}(\varepsilon^2). \tag{5.43}$$

5.6.8 Effective stress-energy from constraint strain

The combination of the mass–strain identification ($\rho_{\text{eff}}c^2 = \Lambda_*\mathcal{I}$), the channel continuity law ($u^\mu\nabla_\mu\mathcal{I} = \theta$), and standard perfect-fluid energy balance (5.41) strictly implies a specific equation of state for the strain sector when $\theta \neq 0$:

$$p = -(\rho_{\text{eff}}c^2 + \Lambda_*) = -\Lambda_*(\mathcal{I} + 1). \tag{5.44}$$

Interpretation. In the constraint-manifold micro-model, mass is directional restriction, which inherently induces a tension-like response of the substrate under adiabatic channel conservation ($\Omega V = \text{const}$). Equation (5.44) makes this explicit: the constant term $-\Lambda_*$ behaves as an effective cosmological constant contribution, while the strain-dependent term contributes additional negative pressure. More general conservation laws $\Omega V^\gamma = \text{const}$ can modify this equation of state and will be explored elsewhere. In the weak-field, nonrelativistic regime evaluated here, only the energy-density component is required to recover Newtonian gravity.

5.6.9 Weak-field limit and Poisson form (consistency check)

In the nonrelativistic, weak-field regime, take $u^\mu \approx (c, 0, 0, 0)$ and $T_{00} \approx \rho_{\text{eff}}c^2$. The covariant strain-balance equation (Eq. (5.3)) then reduces to the Poisson form

$$\nabla^2\Phi = 4\pi G\,\rho_{\text{eff}}, \tag{5.45}$$

with ρ_{eff} determined by the constraint deficit ε via Eq. (5.43). This provides a direct micro-to-macro bridge: *the gravitational source is the local missing solid angle of null-directional support.*

Scope of the metric closure. The present construction establishes the kinematic-to-dynamic bridge at the level of weak-field closure and covariant consistency. A full microscopic derivation of the nonlinear Einstein equations from the informational ensemble remains an open extension.

6 Gauge Forces from the Spinor Representation

The "directional statistics" framework uses S^2 as the space of resolved directions. However, a *complex probability amplitude* over S^2 is most naturally represented on the *spinor lift* of S^2. Concretely, any direction $\hat{\mathbf{k}} \in S^2$ can be represented by a normalized spinor $\psi \in \mathbb{C}^2$ (defined up to phase) via the Bloch/Hopf map

$$\hat{\mathbf{k}} = \psi^\dagger\boldsymbol{\sigma}\,\psi, \qquad \psi^\dagger\psi = 1, \tag{6.1}$$

where $\boldsymbol{\sigma}$ are the Pauli matrices. This representation is one-to-many:

$$\psi \sim e^{i\alpha}\psi \quad \Rightarrow \quad \hat{\mathbf{k}} \text{ unchanged.} \tag{6.2}$$

To compare spinor representatives at nearby spacetime points, one must choose a local section $\psi(\mathbf{x}, t)$ (a phase convention). The canonical $U(1)$ connection associated with this choice is the Berry/Hopf connection

$$\mathcal{A}_\mu(\mathbf{x}, t) \;\equiv\; -\,i\,\psi^\dagger(\mathbf{x}, t)\,\partial_\mu\psi(\mathbf{x}, t). \tag{6.3}$$

Under a local phase change $\psi \mapsto e^{i\alpha(\mathbf{x},t)}\psi$, it transforms as

$$\mathcal{A}_\mu \mapsto \mathcal{A}_\mu + \partial_\mu\alpha, \tag{6.4}$$

so a covariant derivative

$$D_\mu = \partial_\mu - iq\,\mathcal{A}_\mu \tag{6.5}$$

is required for phase-independent comparison of spinor amplitudes. For geometric treatments of gauge connections and fiber bundles, see e.g. [19, 20].

Therefore an intrinsic $U(1)$ redundancy (phase) is *kinematically mandatory* once directions are described by complex amplitudes through the spinor lift.

Let the directional amplitude be written as a lifted complex amplitude on directions, $\eta(\hat{\mathbf{k}})$, so that the state takes the form

$$\Psi = \int_{S^2} \eta(\hat{\mathbf{k}}) \left|\hat{\mathbf{k}}\right\rangle \, d\Omega. \tag{6.6}$$

A local change of spinor phase convention is then a gauge transformation

$$\eta(\hat{\mathbf{k}}; \mathbf{x}, t) \mapsto e^{iq\,\alpha(\mathbf{x},t)} \eta(\hat{\mathbf{k}}; \mathbf{x}, t), \tag{6.7}$$

which necessitates a local covariance connection $\mathcal{A}_\mu$ defined through

$$D_\mu = \partial_\mu - iq\mathcal{A}_\mu, \qquad \mathcal{A}_\mu \mapsto \mathcal{A}_\mu + \partial_\mu \alpha. \tag{6.8}$$

More generally, if directional resolution (or directional correlations) is represented by an N-component internal lift $\eta(\hat{\mathbf{k}}) \in \mathbb{C}^N$, then a local change of internal frame takes the form

$$\eta(\hat{\mathbf{k}}; \mathbf{x}, t) \mapsto U(\mathbf{x}, t)\, \eta(\hat{\mathbf{k}}; \mathbf{x}, t), \qquad U(\mathbf{x}, t) \in U(N). \tag{6.9}$$

When desired, one may factor $U(N) \simeq (SU(N) \times U(1))/\mathbb{Z}_N$ and treat the overall $U(1)$ separately, leaving an $SU(N)$ connection for the non-Abelian sector. Covariance then requires a non-Abelian connection $\mathcal{A}_\mu$ and covariant derivative

$$D_\mu = \partial_\mu - ig\,\mathcal{A}_\mu, \tag{6.10}$$

with gauge transformation law

$$\mathcal{A}_\mu \mapsto U\mathcal{A}_\mu U^{-1} - \frac{i}{g}(\partial_\mu U)U^{-1}, \tag{6.11}$$

and curvature

$$\mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - ig[\mathcal{A}_\mu, \mathcal{A}_\nu]. \tag{6.12}$$

Corollary 6.1 (Gauge structure as kinematic frame tracking)**.** *Gauge fields need not be postulated as independent entities: they arise as the connections required to compare the local internal frame associated with directional spinor amplitudes.*

Scope statement. This establishes the *inevitability* of a $U(1)$ gauge connection from the spinor-lift description of directional amplitudes. Upon generalization to an N-component internal lift, the same covariance logic naturally yields $U(N)$, or equivalently $SU(N)$ together with an overall $U(1)$ sector. A full Standard Model completion still requires selecting the internal state space, symmetry-breaking pattern, and matter representations (charges and chirality) [21].

7 Emergent Macroscopic Dynamics: Gravity and Electromagnetism

A defining test of any foundational substrate theory is its ability to naturally recover the established macroscopic field equations of classical physics without introducing ad hoc forces. In standard quantum field theory, gauge fields and the gravitational metric are treated as distinct, fundamental entities injected into the vacuum.

In the null-directional framework, this distinction vanishes. Both gravitation and electromagnetism emerge through the exact same ontological mechanism: geometric bookkeeping. Because the substrate's local state is defined by a probability distribution of null directions, the local null structure admits descriptions that are redundant under frame choice—specifically, an $SO(1,3)$ tetrad to map the macroscopic light-cones, and a $U(1)$ phase to map the internal spinor lift of the Hopf fibration.

Because these local choices are physically unobservable redundancies, comparing substrate states across the transport manifold requires covariant derivatives. This mathematically mandates the existence of connection fields. In particular, any intrinsic stochasticity of the substrate must enter as fluctuations of these connections (and therefore through their holonomies/curvatures), rather than as ad hoc noise added directly to coordinate-dependent fields. By applying standard effective field theory closures (locality, diffeomorphism and gauge invariance, and leading-order derivatives) to these connections, the classical field equations emerge as natural geometric closures.

Notation Summary. To maintain clarity across macroscopic limits: $\mathcal{A}_\mu$ denotes the microscopic/kinematic connection induced by the spinor lift, A_μ denotes its macroscopic effective gauge field, and $\mathbf{v}$ is the nonrelativistic transport parameterization of the relevant connection component in a chosen frame.

7.1 Emergent Gravity from Null-Frame Redundancy

This subsection provides an independent geometric derivation of the Einstein closure from local Lorentz redundancy. It complements, rather than replaces, the strain-balance construction of Sec. 5.

Theorem 7.1 (Emergent Gravity from Null-Frame Redundancy)**.** *Let the local kinematic state of the null-directional substrate determine a field of admissible null directions at each spacetime point x. Assume this local null structure admits an effective macroscopic description by an orthonormal tetrad (coframe) field $e^a{}_\mu(x)$, defined only up to local Lorentz frame choice,*

$$e^a{}_\mu(x) \;\mapsto\; \Lambda^a{}_b(x)\, e^b{}_\mu(x), \qquad \Lambda(x) \in SO(1,3). \tag{7.1}$$

Then a Lorentz connection $\omega_\mu{}^{ab}(x)$ is required to compare local null frames under transport, and its curvature $R_{\mu\nu}{}^{ab}(\omega)$ supplies the notion of gravitational curvature. Moreover, to leading order in derivatives, imposing locality, diffeomorphism invariance, and second-order field equations (Lovelock's theorem in four dimensions) yields the Einstein field equations (up to a cosmological constant term).

Proof. (1) Null structure ⇒ tetrad and metric. Assume the substrate's local null-directional statistics define an effective local Minkowski structure in each tangent space, represented by a coframe $e^a{}_\mu(x)$ (with a an internal Lorentz index). The emergent metric is

$$g_{\mu\nu}(x) = \eta_{ab}\, e^a{}_\mu(x)\, e^b{}_\nu(x), \tag{7.2}$$

which is invariant under the local Lorentz redundancy (7.1).

(2) Local Lorentz redundancy ⇒ Lorentz connection. Because $e^a{}_\mu(x)$ is defined only up to a local Lorentz choice, comparing frames at neighboring points requires a derivative operator covariant under (7.1). This introduces a Lorentz connection $\omega_\mu{}^{ab}(x)$ and a covariant derivative acting on internal indices:

$$\nabla_\mu V^a \equiv \partial_\mu V^a + \omega_\mu{}^a{}_b\, V^b, \tag{7.3}$$

with transformation law

$$\omega_\mu \;\mapsto\; \Lambda\, \omega_\mu\, \Lambda^{-1} - (\partial_\mu \Lambda)\Lambda^{-1}, \tag{7.4}$$

so that $\nabla_\mu V^a$ transforms covariantly.

In the torsion-free, metric-compatible sector, this connection is uniquely determined by the tetrad (the Levi–Civita spin connection).

(3) Curvature as field strength. The curvature (field strength) of $\omega_\mu{}^{ab}$ is

$$R_{\mu\nu}{}^{ab}(\omega) = \partial_\mu \omega_\nu{}^{ab} - \partial_\nu \omega_\mu{}^{ab} + \omega_\mu{}^a{}_c\, \omega_\nu{}^{cb} - \omega_\nu{}^a{}_c\, \omega_\mu{}^{cb}, \tag{7.5}$$

which measures the failure of parallel transport to be path-independent and encodes gravitational curvature.

(4) Effective closure ⇒ Einstein dynamics. To obtain macroscopic dynamics, impose locality, diffeomorphism invariance, and second-order field equations for the metric degrees of freedom. In four dimensions, Lovelock's theorem implies that the unique lowest-derivative gravitational action satisfying these requirements is the Einstein–Hilbert action (up to a cosmological constant):

$$S_{\text{grav}} = \frac{1}{2\kappa} \int d^4x\, \sqrt{-g}\, (R - 2\Lambda). \tag{7.6}$$

Including the effective action of localized substrate constraints (matter), the total macroscopic action is

$$S_{\text{total}} = S_{\text{grav}} + S_{\text{matter}}. \tag{7.7}$$

Variation of S_{total} with respect to the metric $g_{\mu\nu}$ (or equivalently the tetrad $e^a{}_\mu$ in the torsion-free sector) yields

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa\, T_{\mu\nu}, \tag{7.8}$$

where the stress-energy tensor is defined by

$$T_{\mu\nu} \equiv -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{matter}}}{\delta g^{\mu\nu}}. \tag{7.9}$$

□

Physical Interpretation. Under this theorem, gravity is not a fundamental force field injected into the vacuum. Rather, it is the curvature of the $SO(1,3)$ connection required to compare local null frames defined by the directional substrate. The Einstein equations enter as the unique leading-order covariant effective dynamics of that emergent geometry at macroscopic scales (up to a cosmological term).

7.2 Emergent Electromagnetism from Spinor Phase Redundancy

Theorem 7.2 (Gauge Connection from Spinor Phase Redundancy)**.** *Let the local null-directional state be lifted to a complex 2-spinor amplitude $\psi(x)$. If the physical directional distribution is invariant under local $U(1)$ phase transformations,*

$$\psi(x) \to e^{iq\alpha(x)}\psi(x),$$

then comparison of spinor states at neighboring spacetime points requires the introduction of a $U(1)$ connection A_μ transforming as

$$A_\mu \to A_\mu + \partial_\mu \alpha.$$

Imposing locality, Lorentz covariance, and gauge invariance at leading derivative order, the standard leading-derivative, local, gauge-invariant effective action for this connection in four dimensions is the Maxwell action.

Here the macroscopic gauge field A_μ corresponds to the effective long-wavelength limit of the substrate connection $\mathcal{A}_\mu$ introduced in Section 6.

Proof. This lift is geometrically natural because the celestial sphere of null directions S^2 admits a spinorial representation via the Hopf fibration $S^3 \to S^2$, where a normalized complex 2-spinor defines a direction modulo a $U(1)$ phase. Thus the local $U(1)$ phase redundancy is not imposed, but arises from the intrinsic fiber structure of the null-directional state space.

Because this overall phase is physically unobservable, comparing directional states at neighboring spacetime points requires a derivative that is covariant under local phase choices:

$$D_\mu \psi = (\partial_\mu - iqA_\mu)\psi.$$

Covariance under $\psi \to e^{iq\alpha}\psi$ mandates that the connection transform as

$$A_\mu \to A_\mu + \partial_\mu \alpha.$$

The gauge-invariant curvature of this connection,

$$F_{\mu\nu} \equiv \frac{i}{q}[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu,$$

automatically satisfies the Bianchi identity

$$\partial_{[\lambda} F_{\mu\nu]} = 0,$$

which corresponds to the homogeneous Maxwell equations.

To establish macroscopic dynamics, impose locality, Lorentz covariance, and gauge invariance. The unique lowest-derivative action satisfying these requirements is

$$S = \int d^4x \left(-\frac{1}{4} F_{\mu\nu}F^{\mu\nu} + A_\mu J^\mu \right).$$

Stationarity of this action yields

$$\partial_\mu F^{\mu\nu} = J^\nu.$$

Taking the divergence implies $\partial_\mu J^\mu = 0$, so charge conservation follows as a consistency condition of gauge invariance. □

Scope clarification. This argument establishes the kinematic inevitability of gauge connections from internal frame redundancy in the spinor lift of directional amplitudes. It does not by itself determine the observed electric charge assignments, matter representations, or symmetry-breaking pattern of the Standard Model. Those require additional structure beyond the minimal null-directional substrate.

Physical Interpretation. Under this theorem, the electromagnetic field is not a fundamental substance injected into the vacuum. Rather, A_μ is the geometric connection required to continuously synchronize the internal spinor phases of the null-directional ensemble across the transport manifold.

Phase from connection transport (holonomy template)

A unifying feature of the connection viewpoint is that interferometric phases are path functionals of a connection. Schematically, for a matter-wave arm with 4-momentum p_μ transported along a path γ, the phase can be written as an action in units of $\hbar$,

$$\phi[\gamma] \;=\; \frac{1}{\hbar}\int_\gamma p_\mu\, dx^\mu \;+\; \phi_{\rm conn}[\gamma], \tag{7.10}$$

where $\phi_{\rm conn}$ denotes additional geometric phase contributions arising from the relevant connection(s) that compare internal frames along transport (e.g. the $U(1)$ connection for phase and the Lorentz connection for local frames). Consequently, if the connection contains a fluctuating component, the induced phase perturbation is linear in the arm momentum scale, and its variance is generically momentum-quadratic:

$$\delta\phi \;\propto\; \frac{p_{\rm arm}}{\hbar}\int dt\, \delta\mathcal{A}(t) \qquad \Longrightarrow \qquad \langle \delta\phi^2\rangle \;\propto\; \left(\frac{p_{\rm arm}}{\hbar}\right)^2. \tag{7.11}$$

This is the structural origin of the $p_{\rm arm}^2$ (and, in transport-defined regimes, m^2) amplification used below.

8 Prediction: A Kinematic Coherence Bound

In addition to deterministic shear bounds, the stochastic null-directional hypothesis predicts a quadratic mass-dependent visibility decay characterized by an effective saturation parameter $\xi_{\rm eff}$. This section isolates the key test-facing statement: a bound linking mass, flow gradients, and coherent spatial support.

8.1 Transport coupling (adopted minimal form)

The standard moving-background (transport-covariant) coupling is adopted as

$$\hat{H} = \frac{1}{2m}\left(-i\hbar\nabla - m\mathbf{v}(\mathbf{x},t)\right)^2, \tag{8.1}$$

interpreting $\mathbf{v}$ as the first moment (3.1). (Deriving (8.1) from an underlying null microdynamics is deferred.) For spatially varying $\mathbf{v}(\mathbf{x},t)$ interpret (8.1) as the standard Hermitian operator with $\hat{\mathbf{p}} = -i\hbar\nabla$ acting on everything to its right; equivalently one may use the symmetrized form $-\frac{1}{2}\big(\mathbf{v}\cdot\hat{\mathbf{p}} + \hat{\mathbf{p}}\cdot\mathbf{v}\big)$ for the cross term. In the upgraded connection viewpoint, $\mathbf{v}$ is interpreted as a convenient nonrelativistic parameterization of the relevant transport connection in a chosen frame; any intrinsic stochasticity is therefore attributed to fluctuations of the underlying connection (and hence of transport), not to an independently postulated noise source.

8.2 Gradient-induced momentum shear

Let $G_{ij} \equiv \partial_j v_i$ and let $|\nabla \mathbf{v}|$ denote a characteristic shear magnitude (e.g. the operator norm or RMS norm of G). For a wavepacket of spatial width L in a region with $\nabla \mathbf{v} \neq 0$,

$$\Delta p_{\text{flow}} \sim m\,|\nabla \mathbf{v}|\,L. \tag{8.2}$$

Fourier duality gives an intrinsic spread for spatial width L,

$$\Delta p_{\text{qm}} \gtrsim \frac{\hbar}{2L}. \tag{8.3}$$

A minimal coherence criterion is $\Delta p_{\text{flow}} \lesssim \Delta p_{\text{qm}}$, yielding

$$L^2 \lesssim \frac{\hbar}{2m|\nabla \mathbf{v}|}. \tag{8.4}$$

Define the characteristic coherence scale (up to order-unity shape factors):

$$\boxed{L_{\text{coh}} \sim \sqrt{\frac{\hbar}{2m|\nabla \mathbf{v}|}} \quad = \quad \mathcal{O}(1) \times \sqrt{\frac{\hbar}{m|\nabla \mathbf{v}|}}.} \tag{8.5}$$

8.3 Weak-field estimate

For stationary radial Newtonian flow (up to factors of order unity depending on how the gradient magnitude is defined) $|\mathbf{v}(r)| = \sqrt{2GM/r}$,

$$|\nabla \mathbf{v}| \sim \sqrt{\frac{GM}{r^3}}, \tag{8.6}$$

so

$$L_{\text{coh}} \sim \sqrt{\frac{\hbar}{m}} \left(\frac{r^3}{GM}\right)^{1/4}. \tag{8.7}$$

Interpretation. Equation (8.5) organizes a familiar fact (tidal gradients produce phase shear) into a single compact scaling that directly links the first-moment transport field and quantum support. It is therefore a clean place to anchor experimental discussion.

Scope of the bound. Equation (8.5) is a kinematic coherence organizer conditional on a specified transport field $\mathbf{v}(\mathbf{x}, t)$. By itself it does not distinguish standard GR+QM from the null-ensemble hypothesis, since deterministic gravity gradients also induce phase shear across extended states. The discriminator enters only if $\mathbf{v}$ contains an irreducible stochastic component with definite correlation scales. The following section isolates that distinction explicitly.

9 Deterministic Tidal Shear vs. Stochastic Directional Jitter

Two mechanisms can reduce interferometric visibility in gravitational environments:

1. **Deterministic tidal shear** (GR+QM baseline): coherent phase curvature across an extended state in a gravity gradient; in principle refocusable.
2. **Stochastic directional jitter** (candidate discriminator): irreducible random fluctuations of the transport field beyond the classical mean, producing diffusive phase noise and irreversible decoherence.

9.1 Deterministic tidal shear (baseline)

In standard GR+QM (weak-field, nonrelativistic),

$$\hat{H}_{\mathrm{GR+QM}} = \frac{\hat{\mathbf{p}}^2}{2m} + m\Phi(\mathbf{x}), \tag{9.1}$$

and expanding Φ to second order introduces the tidal tensor $\Gamma_{ij} = \partial_i \partial_j \Phi$. Let Γ denote a characteristic magnitude of the tidal tensor (e.g. an eigenvalue scale). An order-of-magnitude tidal phase shear across extent L accumulated over time t is

$$\Delta\phi_{\mathrm{tide}} \sim \frac{m}{\hbar}\left(\frac{1}{2}\,\Gamma\,L^2\right) t. \tag{9.2}$$

9.2 Stochastic transport jitter as connection fluctuations (Planck-anchored minimal model)

The null-directional framework identifies classical fields as connection bookkeeping: a $U(1)$ connection for internal phase and an $SO(1,3)$ (spin) connection for local null frames. Accordingly, any intrinsic "substrate jitter" is modeled as a fluctuating component of the relevant transport connection. In the weak-field, nonrelativistic limit, this appears as fluctuations of the effective transport field entering the matter-wave phase.

The transport connection is written schematically as

$$\mathcal{A} \;=\; \bar{\mathcal{A}} \;+\; \delta\mathcal{A}, \qquad \langle \delta\mathcal{A} \rangle = 0, \tag{9.3}$$

where $\mathcal{A}$ denotes the effective connection component that controls the relative phase accumulated between interferometer arms (in practice, a particular projection of the underlying $U(1)$ and/or Lorentz transport connection along the arms).

Phase diffusion and momentum-quadratic scaling. Because interferometric phase is an action in units of $\hbar$, a small fluctuating transport contribution produces a phase perturbation linear in the arm momentum scale. Here, the effective arm-projected connection component $\mathcal{A}$ carries kinematic dimensions of velocity (L/T), consistent with its nonrelativistic interpretation as a fluctuating transport flow. Consequently, the integral $\int \delta\mathcal{A}\,dt$ carries dimensions of length, yielding:

$$\delta\phi \;\sim\; \frac{p_{\mathrm{arm}}}{\hbar}\int_0^T dt\,\delta\mathcal{A}_{\mathrm{diff}}(t), \tag{9.4}$$

where $\delta\mathcal{A}_{\mathrm{diff}}$ is the differential fluctuation between arms. For stationary fluctuations with finite correlation time τ_c, long-time diffusion implies

$$\Gamma_\phi \;\sim\; \left(\frac{p_{\mathrm{arm}}}{\hbar}\right)^2 \langle \delta\mathcal{A}_{\mathrm{diff}}^2 \rangle\, \tau_c, \tag{9.5}$$

recovering the structural p_{arm}^2 amplification (and hence the m^2 regime when $p_{\mathrm{arm}} \propto m$).

Planck-anchored diffusion scale. Rather than introducing independent microscopic parameters, the fluctuation strength is compressed into a single Planck-anchored diffusion combination. Let the scalar directional strain

$$\mathcal{I}(x) \equiv \ln\!\left(\frac{4\pi}{\Omega(x)}\right) \tag{9.6}$$

serve as the local control parameter. It is postulated that the minimal diffusion combination

$$\boxed{\mathcal{D}_{\rm null}(x) \;\equiv\; \langle \delta\mathcal{A}_{\rm diff}^2 \rangle\, \tau_c \;=\; \xi\, c\ell_P\, \mathcal{I}(x),} \tag{9.7}$$

where $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length and ξ is a dimensionless fundamental amplitude parameter. Experimental observables constrain an effective saturation parameter $\xi_{\rm eff}$ that bundles ξ with geometry and interrogation time.

Combining Eqs. (9.5) and (9.7) gives the minimal decoherence rate template

$$\boxed{\Gamma_\phi(x, L) = \left(\frac{p_{\rm arm}}{\hbar}\right)^2 \xi\, c\ell_P\, \mathcal{I}(x)\, \mathcal{F}\!\left(\frac{L}{\ell_c(x)}\right),} \tag{9.8}$$

where $\mathcal{F}(y) = 1 - e^{-y}$ encodes baseline-controlled turn-on.

This construction ensures:

- vanishing stochasticity in isotropic vacuum ($\mathcal{I} \to 0$),
- proportional growth in stronger strain environments,
- a strict conceptual separation of scales: the *amplitude* of the diffusion is anchored exclusively by the fundamental Planck scale ℓ_P, while the *spectral shape* (and the turnover frequency in Eq. (9.12)) is governed by the independent correlation parameter τ_c.

Correlation scale. The spatial correlation length is tied to curvature rather than an environmental parameter. The only available covariant scalar length constructed from curvature is

$$\ell_c(x) \sim \left| R_{\alpha\beta\gamma\delta} R^{\alpha\beta\gamma\delta} \right|^{-1/4}. \tag{9.9}$$

In weak spherical fields this reduces parametrically to

$$\ell_c(r) \sim \sqrt{\frac{r^3}{GM}}, \tag{9.10}$$

so decorrelation is governed by tidal curvature rather than laboratory geometry.

Spectral signature. If stationary Gaussian fluctuations with exponential temporal correlation

$$\langle \delta\mathcal{A}_{\rm diff}(t)\, \delta\mathcal{A}_{\rm diff}(0) \rangle \propto e^{-|t|/\tau_c} \tag{9.11}$$

are assumed, the phase-noise power spectral density is Lorentzian. Here Γ_ϕ is defined by the long-time diffusive visibility law $V(T) \simeq \exp(-\Gamma_\phi T)$, which fixes the normalization convention for S_ϕ:

$$\boxed{S_\phi(f) = \frac{2\Gamma_\phi}{(2\pi f)^2 + (1/\tau_c)^2}.} \tag{9.12}$$

An intrinsic turnover frequency $f_c \sim 1/(2\pi\tau_c)$ is therefore predicted, distinguishing this mechanism from the typically low-frequency dominated spectrum of terrestrial gravity-gradient noise.

Separation from Newtonian noise. Unlike classical gravity-gradient noise, which scales linearly with tidal tensor components and correlates with environmental density fluctuations, the present mechanism predicts:

1. quadratic scaling with interferometer arm momentum ($\Gamma_\phi \propto p_{\rm arm}^2$),
2. proportionality to intrinsic directional strain ($\propto \mathcal{I}(x)$),
3. baseline-controlled onset governed by curvature ($L/\ell_c(x)$),
4. spectral turnover at f_c independent of environmental subtraction procedures.

Alternative models of gravitationally induced decoherence include those of Diósi and Penrose, as well as stochastic gravity approaches [22–26].

Mean-strain floor vs. fluctuation-driven Newtonian noise. A qualitative separation from terrestrial Newtonian (gravity-gradient) noise is obtained by distinguishing *mean-field proportionality* from *fluctuation-driven backgrounds*. In particular, the strain-linked template predicts persistence under subtraction of correlated environmental monitors, whereas Newtonian noise is reducible by such procedures in principle. In the present template, the diffusion amplitude is proportional to the local *mean* directional strain $\mathcal{I}(x) = \ln(4\pi/\Omega)$, so a nonzero diffusion floor is predicted even in the idealized limit where classical environmental density fluctuations are absent. By contrast, Newtonian noise arises from *fluctuations* in the gravitational field sourced by time-dependent density perturbations (seismic and atmospheric), and therefore must track environmental monitors and site conditions, admitting subtraction using correlated sensors and geometry. Accordingly, residual decoherence that persists under aggressive environmental subtraction and that scales with the static strain proxy $\mathcal{I}(x)$ (rather than with measured density-fluctuation spectra) provides the intended discriminator against Newtonian-noise explanations.

Terrestrial benchmark. At Earth's surface, $\mathcal{I}_\oplus \approx 1.4 \times 10^{-9}$. With $c\ell_P \approx 4.85 \times 10^{-27}\,\mathrm{m}^2/\mathrm{s}$, the diffusion coefficient is

$$\mathcal{D}_{\mathrm{null},\oplus} \sim \xi \times 10^{-36}\,\mathrm{m}^2/\mathrm{s}. \tag{9.13}$$

For recoil-defined momenta typical of current long-baseline atom interferometers, the resulting $\xi_{\rm eff}$ implied by null results corresponds to critical mass scales well above current interferometric masses. Terrestrial measurements therefore provide clean null-test constraints on ξ without conflicting with existing data.

To contextualize the experimental horizon: for a proposed 10^6 amu transition-scale target (e.g., a levitated nanoparticle or virus) with an arm separation baseline $L \sim 1$ m and coherence time $T \sim 1$ s, the predicted momentum-quadratic phase diffusion would emerge distinctly from standard environmental backgrounds if the saturation parameter approaches $\xi_{\rm eff} \sim 10^{-12}$. For fluctuation amplitudes ξ below current bounds, this lies strictly at or just below the sensitivity horizon of next-generation macroscopic matter-wave interferometers, providing a concrete near-term target.

Corresponding bound. For fixed interferometer parameters $(p_{\rm arm}, L, T)$, null spin-echo results imply

$$\xi \lesssim \frac{1}{(p_{\rm arm}/\hbar)^2\, c\ell_P\, \mathcal{I}(x)\, T} \ln\!\left(\frac{1}{V_{\rm echo}}\right). \tag{9.14}$$

A sufficiently small bound on ξ implies that any Planck-scale null-directional stochasticity is strongly suppressed in the tested strain regime.

9.3 Spin-Echo Discriminator: Reversibility vs. Irreversibility

A decisive experimental discriminator between deterministic tidal shear (GR+QM baseline) and stochastic transport fluctuations is provided by a time-reversal ("spin-echo") interferometer sequence.

Deterministic baseline (refocusable). In standard GR+QM, gravity gradients introduce deterministic phase curvature. For a Mach–Zehnder interferometer with arm separation $L(t)$, a gravity gradient produces a phase shift proportional to the tidal tensor (Eq. (9.2)). In idealized GR+QM (absent environmental and technical noise), the static tidal contribution is, in principle, refocusable under a time-symmetric echo sequence. Thus

$$V_{\text{echo}}^{(\text{GR})} \text{ is limited only by known technical and environmental decoherence channels.} \tag{9.15}$$

Deterministic tidal shear is therefore, in principle, reversible.

Stochastic transport fluctuations (irreversible). Suppose instead that the relevant transport connection contains a fluctuating component

$$\mathcal{A} = \bar{\mathcal{A}} + \delta\mathcal{A}, \qquad \langle \delta\mathcal{A} \rangle = 0, \tag{9.16}$$

with finite correlation time τ_c and spatial correlation length ℓ_c. The differential fluctuation between interferometer arms is

$$\delta\mathcal{A}_{\text{diff}}(t) \equiv \delta\mathcal{A}(\mathbf{x}_1, t) - \delta\mathcal{A}(\mathbf{x}_2, t). \tag{9.17}$$

Assuming isotropic stationary fluctuations,

$$\left\langle \delta\mathcal{A}_{\text{diff}}^2 \right\rangle = 2\sigma_{\mathcal{A}}^2 \left[1 - C\left(\frac{L}{\ell_c} \right) \right], \tag{9.18}$$

where $C(r)$ is the normalized spatial correlation function.

Phase diffusion under transport coupling. Under the transport Hamiltonian

$$\hat{H} = \frac{1}{2m} \left(-i\hbar\nabla - m\mathbf{v} \right)^2, \tag{9.19}$$

connection fluctuations induce phase diffusion while the effective arm momentum scale p_{arm} is set by the interferometer geometry (e.g. $p_{\text{arm}} \sim m v_{\text{arm}}$ in transport-defined setups or $p_{\text{arm}} \sim N\hbar k_{\text{eff}}$ in LMT interferometers). Here $\delta\mathcal{A}_{\text{diff}}$ denotes the effective arm-projected fluctuation of the transport connection entering the phase in the weak-field, nonrelativistic limit. Up to an order-unity kinematic factor set by the interferometer geometry, this reduces to the same momentum-quadratic (p_{arm}^2) scaling as Eq. (9.8). For interrogation times $T \gg \tau_c$,

$$\Gamma_\phi \sim \left(\frac{p_{\text{arm}}}{\hbar} \right)^2 \left\langle \delta\mathcal{A}_{\text{diff}}^2 \right\rangle \tau_c. \tag{9.20}$$

Up to an order-unity geometry factor encoded in p_{arm}, this is consistent with the minimal diffusion scaling in Eq. (9.8).

Effective saturation parameter. Writing the visibility in Gaussian diffusion form,

$$V(T) = V_0 \exp[-\Phi], \tag{9.21}$$

with accumulated phase variance $\Phi = \Gamma_\phi T$, the quadratic-mass regime motivates the parameterization

$$\Phi = \xi_{\text{eff}}\, m^2. \tag{9.22}$$

This defines the *effective saturation parameter*

$$\xi_{\text{eff}} \equiv \frac{\Gamma_\phi T}{m^2}, \tag{9.23}$$

which is experiment-dependent and bundles ξ, T, τ_c, $\mathcal{I}(x)$, and geometry.

Momentum scaling and the m^2 regime. The interferometric phase is an action in units of $\hbar$, $\phi = S/\hbar$. Under the transport coupling, the phase perturbation from a small fluctuation is schematically $\delta\phi \sim \frac{1}{\hbar}\int p_{\text{arm}}\, \delta\mathcal{A}_{\text{diff}}(t)\, dt$, so $\delta\phi$ is *linear* in the arm momentum $p_{\text{arm}} \propto m$. Consequently the phase variance (and hence the diffusion/decoherence rate) scales quadratically:

$$\left\langle \delta\phi^2 \right\rangle \propto p_{\text{arm}}^2, \qquad \Rightarrow \qquad \Gamma_\phi \propto p_{\text{arm}}^2. \tag{9.24}$$

In transport-defined regimes ($p_{\text{arm}} \sim m v_{\text{arm}}$), this implies quadratic mass scaling,

$$\Gamma_\phi \propto m^2. \tag{9.25}$$

This motivates heavy-mass superpositions as the most sensitive probes, and provides a clean discriminant against many surface-interaction noise sources whose rates scale primarily with cross-section, polarizability, or flux rather than m^2. Quadratic mass amplification follows only in configurations where $p_{\text{arm}} \sim m v_{\text{arm}}$.

The resulting spin-echo visibility is

$$V_{\text{echo}}(T) \simeq \exp(-\Gamma_\phi T)\,. \tag{9.26}$$

Key discriminator. Unlike deterministic tidal shear, stochastic phase diffusion cannot be refocused. The null-directional ensemble therefore predicts:

1. **Momentum-quadratic amplification:** The decoherence rate scales quadratically with arm momentum, modulated by the local strain:

$$\Gamma_\phi \propto p_{\text{arm}}^2\, \mathcal{I}(x). \tag{9.27}$$

   In transport-defined regimes ($p_{\text{arm}} \sim m v_{\text{arm}}$), this yields quadratic mass amplification. In recoil-defined regimes ($p_{\text{arm}} \sim N\hbar k_{\text{eff}}$), the scaling is quadratic in the effective recoil momentum.

2. **Baseline turn-on:** visibility loss increases once $L \gtrsim \ell_c$.

3. **Irreversibility:** after an ideal spin-echo sequence,

$$V_{\text{echo}}^{(\text{stat})} < V_{\text{echo}}^{(\text{GR})}, \tag{9.28}$$

   where $V_{\text{echo}}^{(\text{GR})}$ is limited only by known technical and environmental decoherence channels.

4. **Spectral/mitigation signature (vs. Newtonian noise).** Terrestrial "Newtonian noise" (gravity-gradient noise from seismic and atmospheric density fluctuations) is typically dominated by low frequencies and can be reduced by environmental monitoring, subtraction, and site/geometry choices. By contrast, stochastic transport jitter in the null-ensemble picture is *intrinsic* and characterized by a finite correlation time τ_c (and length ℓ_c), implying a distinctive turnover near $f \sim 1/\tau_c$ and persistence under mitigation strategies that remove low-frequency environmental gravity-gradient backgrounds.

Interpretation. Deterministic gravity gradients correspond to a smooth manifold description and are reversible under time-symmetric control. A residual, momentum-amplified visibility loss after spin-echo would instead indicate an underlying statistical microstructure of transport (with quadratic mass amplification realized only in transport-defined regimes where $p_{\rm arm} \sim m v_{\rm arm}$). Conversely, null results directly constrain the fluctuation amplitude $\sigma_{\mathcal{A}}$ and correlation time τ_c of the null ensemble.

This test cleanly separates:

- deterministic tidal shear (baseline GR+QM),
- from irreducible stochastic transport jitter.

Note on null results (parameter bounds). If a high-fidelity spin-echo sequence restores visibility to unity up to experimental systematics, then Eq. (9.26) implies an upper bound on the effective diffusion strength, schematically

$$\Gamma_\phi \lesssim \frac{1}{T} \ln\left(\frac{1}{V_{\rm echo}}\right), \tag{9.29}$$

and therefore (using Eqs. (9.20) and (9.18)) a direct constraint on the product $\sigma_{\mathcal{A}}^2 \tau_c$ (modulo the correlation factor $1 - C(L/\ell_c)$ and the known interferometer momentum $p_{\rm arm}$). Such bounds would quantify how closely the statistical transport field approximates a smooth manifold on the mass–time–baseline scales probed by the experiment.

Defining a critical mass scale m_* (using $\Phi = \xi_{\rm eff} m^2$ as defined above) by the condition $\Phi(m_*) = 1$, one obtains

$$\xi_{\rm eff} = \frac{1}{m_*^2}. \tag{9.30}$$

The mass scale m_* therefore provides a convenient experiment-facing proxy for the effective saturation strength.

Explicit falsifiability statement. For fixed interferometer parameters $(p_{\rm arm}, L, T)$, the strain-linked benchmark predicts a definite, environment-dependent diffusion floor parameterized by ξ through Eq. (9.8). Improved spin-echo bounds therefore directly constrain

$$\xi \lesssim \frac{1}{(p_{\rm arm}/\hbar)^2\, c\ell_P\, \mathcal{I}(x)\, T} \ln\left(\frac{1}{V_{\rm echo}}\right). \tag{9.31}$$

Separately, using $\xi_{\rm eff} = 1/m_*^2$ (Eq. (9.30)) and the definition $\ln(1/V_{\rm echo}) \simeq \Gamma_\phi T$, the bound may be expressed as

$$m_* \gtrsim \frac{m}{\sqrt{\ln\left(\frac{1}{V_{\rm echo}}\right)}}, \tag{9.32}$$

where m is the *tested* interfered mass in the experiment. In transport-defined regimes where $p_{\rm arm} \simeq m v_{\rm arm}$, Eq. (9.8) implies

$$\xi_{\rm eff}(x, L) \sim \left(\frac{v_{\rm arm}}{\hbar}\right)^2 \xi\, c\ell_P\, \mathcal{I}(x)\, \mathcal{F}\left(\frac{L}{\ell_c(x)}\right) T, \tag{9.33}$$

and therefore $m_* \sim \xi_{\rm eff}^{-1/2}$ for a given configuration.

In the absence of any residual beyond known technical and environmental sources, the parameter ξ is driven toward zero, thereby implying that the null substrate is effectively classical (metric-like) over the probed scales and excluding Planck-strength stochastic transport fluctuations in the tested regime.

10 Discussion: Programmatic Structure and Extensions

The theorem isolates a *single primitive object* (directional statistics of null propagation) from which SR and weak-field gravitational limits (via moments), Fourier-duality constraints on quantum localization, and gauge structure (via internal redundancy) follow through a combination of structural consequences and constitutive closures. What remains for a "GUT-level" completion is not the emergence of gauge *geometry* (already implied by local internal covariance), but the selection of:

- the internal state space and symmetry group (why $SU(3) \times SU(2) \times U(1)$, or its embedding),
- the matter content (representations, chirality, charge assignments),
- and the dynamics selecting symmetry breaking and couplings.

The prediction-facing part of this program is the coherence-gradient scaling (8.5) and the possible presence (or absence) of mass-amplified diffusive decoherence (9.8)–(9.26) beyond deterministic tidal shear (Eq. (9.2)) [27].

Additionally, the kinematic and strain functionals developed here are not confined to local weak-field regimes. In a companion work, the same directional-statistical framework is explored in connection with the statistical origin of the Principle of Least Action [28]. Separately, in [1], the strain formalism is extended to early-universe structure formation, yielding a kinematic capacity envelope for supermassive black hole growth at $z > 10$. That derivation employs the ΛCDM transfer function with Planck 2018 cosmological parameters and leads to a falsifiable prediction for the high-mass tail and redshift scaling of early black hole populations. Together, these companion applications provide an explicit quantitative extension of the present framework beyond the local kinematic limits emphasized here.

11 Conclusion

This work develops a compressed kinematic framework based on invariant null propagation and directional statistics. In one spatial dimension, relativistic clock rate is *constitutively identified* with a symmetric overlap functional of opposing directional weights, from which the Lorentz factor follows exactly within the reduced directional sector without presupposing spacetime geometry. Proper time thereby acquires an operational interpretation as preserved symmetric directional support along a worldline.

Within the same framework, first-moment transport in the stationary, irrotational limit reproduces the Newtonian gravitational potential, while Fourier duality of directional amplitudes

constrains admissible localization in the narrowband angular sector and provides the directional foundation for the standard Heisenberg uncertainty relation. In the unrestricted case, the full relation is recovered through the usual Fourier pair on momentum space. Promoting deviations from isotropy to a covariant scalar strain functional aligns null focusing with metric curvature under standard locality, covariance, and four-dimensional weak-field closure assumptions, yielding the Einstein field equations as the unique leading-order covariant closure once an effective metric description is assumed. Likewise, lifting directional amplitudes to their spinor representation introduces intrinsic internal redundancy, from which gauge connections arise as the minimal structures required for local comparison.

These results are structural rather than revisionary: established field equations are not replaced, but are shown to admit a unified reinterpretation as statistical functionals of an invariant null-directional substrate. In this picture, classical fields represent geometric bookkeeping of comparison obstructions, while relativistic time dilation reflects a statistical limitation on symmetric directional support under the adopted overlap-clock constitutive law.

The coherence bound (8.5) links mass, gradients, and coherent support in a compact form, and the Planck-anchored stochastic transport template (9.8) provides a concrete experimental discriminator between deterministic tidal shear and irreducible directional fluctuations. Whether null-directional statistics represent a deeper microstructure or merely a reformulation of known kinematics remains an open question, but the framework demonstrates that a single invariant null ensemble can organize several core structures of relativistic and quantum theory within a unified statistical language.

Limitations and Open Problems

The present framework does not yet provide a microscopic dynamical law for the evolution of the directional ensemble itself, nor a derivation of Standard Model matter content from first principles. The overlap-clock identification, the weak-field Poisson-type strain closure, and the stochastic transport template are minimal constitutive or effective closures chosen for empirical consistency; their deeper derivation from an underlying null-update microdynamics remains an open problem. These gaps define the next stage of the program.

Statements and Declarations

Competing Interests: The author declares no competing interests.
Data Availability: Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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